

WV - EWP 2025
ELKHORN CREEK
McDOWELL COUNTY, WEST VIRGINIA
DSR 54-02-25-5087-001

STREAMBANK STABILIZATION

LIST OF SPECIFICATIONS

CONSTRUCTION SPECIFICATIONS

NO.	DATE	TITLE
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5	6/24	Pollution Control
6	6/24	Seeding, Sprigging and Mulching
8	6/24	Mobilization and Demobilization
9	6/24	Traffic Control
11	6/24	Removal of Water
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Construction Specification 4 — Channel Clearing and Shaping

Title 210 - Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of clearing designated areas by the removal and disposal of trees, logs, stumps, shrubs, brush, and rubbish and the shaping of the channel.

B. Protection of Existing Vegetation

1. Protect trees and other vegetation designated to remain undisturbed from damage throughout the duration of the construction period. Any damages resulting from the contractor's operations or neglect must be repaired by the contractor.
2. Earthfill, stockpiling of materials, vehicle or equipment parking, and excessive foot or vehicle traffic must not be allowed within the drip line of vegetation designated to remain in place. Vegetation damaged by any of these or similar actions must be replaced with viable vegetation of the same species, similar condition, and like size unless otherwise approved by the contracting officer.
3. Any cuts, skins, scrapes, or bruises to the bark of the vegetation must be carefully trimmed and local nursery-accepted procedures must be used to seal damaged bark.
4. Any limbs or branches 0.5 inch or larger in diameter that are broken, severed, or otherwise seriously damaged during construction must be cut off at the base of the damaged limb or branch, flush with the adjacent limb or tree trunk.
5. All roots 1 inch or larger in diameter that are cut, broken, or otherwise severed during channel shaping must have the ends smoothly cut perpendicular to the root. Roots exposed during channel shaping operations must be covered with moist soil as soon as possible to prevent them from drying out.

C. Marking

1. The limits of the areas to be cleared and shaped are marked by stakes, flags, paint, tree markings, or other suitable methods or as specified in Section H of this specification or will be shown on the drawings. Trees to remain standing, undisturbed, and uninjured are designated by special markings.

D. Clearing

1. Trees and other vegetation marked for clearing must be cut off as near the ground surface as conventional tools and equipment normally permit. All trees not

marked for preservation and all snags, logs, brush, shrubs, stumps, and rubbish must be cleared from within the area limits identified.

E. Disposal

1. Dispose of all woody material, vegetation, and rubbish resulting from clearing from designated areas at the locations and in a manner shown on the drawings or as specified in Section H of this specification.

F. Shaping

1. Shape the channel bottom and side slopes as shown on the drawings. The resulting shaped channel surface must be reasonably smooth. Remove and dispose material excavated during the channel-shaping operation from the channel as specified in Section H of this specification.

G. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established by the contract, the designated cleared and shaped area is measured and area is determined to the nearest 0.1 acre. Payment for clearing and shaping is made at the contract unit price for the item and constitutes full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.
2. Method 2. For items of work for which specific unit prices are established by the contract, the length of the cleared and shaped channel designated is measured to the nearest 100 feet. Payment for clearing and shaping is made at the contract unit price for the item and constitutes full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory performance of the work.
3. Method 3. For items of work for which specific unit prices are established by the contract, the cleared and shaped areas is measured, and area is determined to the nearest 0.1 acre. The designated cleared and shaped areas is determined from the measured width at representative sections and the distance between the sections. Payment for clearing and shaping is made at the contract unit price for the item and constitutes full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.
4. Method 4. For items of work for which specific lump sum prices are established by the contract, the extent of clearing and shaping is not measured or determined for payment. Payment for clearing and shaping is made at the contract lump sum price for the item and constitutes full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.
5. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which

it is made subsidiary. Such items and items to which they have been made subsidiary are identified in Section H of this specification.

H. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Channel Clearing and Shaping, General Requirements

- 1) In Section C, Marking. The construction drawings will designate the ingress and egress and construction limits.
- 2) In Section D, Clearing. It is not intended that live trees, shrubs, brush be removed, unless designated otherwise in the specifications or drawings or designated by the Contracting Officer or Contracting Officer's Representative.
- 3) In Section E, Disposal.
 - i. Burnable materials may be burned in accordance with local and state regulations, if allowed by the Contractor Officer.
 - ii. Non-burnable materials and materials from regulated burning areas, or all materials if burning is not permitted by the Contracting Officer, shall be disposed of offsite and outside of the 100-year floodplain limits. Disposal shall be in accordance with all local, state, and federal regulations. The locations of offsite disposal areas and the manner of disposal will require the approval of the Contractor Officer.
- 4) In Section F, Shaping.
 - i. All deposition removal, channel shaping shall be excavated and filled to the lines and grades designated in the drawings. Deposition removal shall include, but not limited to, the removal of silt, sand, gravel, and boulders. All excavation shall be conducted in a manner which will not restrict flow in the existing channel and shall conform to Construction Specification 21, Excavation. All fill shall be conducted in a manner which will not restrict flow in the existing channel and shall conform to Construction Specification 23, Earthfill and Construction Specification 25, Rockfill.
 - ii. Soil and rock materials from required excavations shall be used to reconstruct the channel as shown on the drawings, as necessary. Debris, rubbish, trees, brush, or any other foreign material shall not be used for channel shaping and shall not be buried in the shaped channel. The reconstructed streambanks shall be shaped to facilitate positive drainage to the channel.
 - iii. Excess soil and rock materials from the channel shaping shall be disposed of at the locations shown on the drawings and as approved by the Contractor Officer. If no disposal area is shown on the drawings, the Contractor shall be responsible for obtaining an offsite disposal area. The disposal area must be located out of the 100-year floodplain. The offsite disposal area must be graded to allow positive drainage and be seeded and mulched as required by Construction Specification 6, Seeding, Sprigging, and Mulching and as required by the property owner. The Contractor shall provide a copy of the written agreement with the property owner to the Contractor Officer.
 - iv. The manner and location of disposal will require approval by the Contractor Officer. Disposal shall be in accordance with all local, state, and federal regulations.

5) Erosion and Pollution Control.

- i. Perform construction operations in accordance with Construction Specification 5, Pollution Control. Operate construction equipment from the channel bank and use only hand-operated equipment in the channel wherever possible.

b. Subsidiary Item, Channel Clearing and Shaping

- 1) This item shall consist of all work required to clear designated areas by the removal and disposal of trees, logs, stumps, shrubs, brush, and rubbish, and the shaping of the channel as defined in the construction specifications and shown on the construction drawings.
- 2) In Section G, Measurement and Payment, shall not apply.
- 3) No specific payment shall be made for Channel Clearing and Shaping. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Grouted Rock Riprap, Construction Specification 62 for Site 1 and Bid Item 5B, Gabions, Construction Specification 64 for Site 2.**

Construction Specification 5 — Pollution Control

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of installing measures or performing work to control erosion and minimize the production of sediment and other pollutants to water and air from construction activities.
2. The following BioPreferred® product categories are applicable to this specification:
 - a. Mulch and compost materials
 - b. Erosion control materials
 - c. Fertilizers
 - d. Dust suppressants
 - e. Agricultural spray adjuvants

B. Material

1. Silt fence conform to the requirement of Materials Specification 592, Geotextile. All other material furnished meet the requirements of the material specifications listed in Section G of this specification.
2. Erosion and Sediment Control Measures and Works. The measures and works includes, but are not limited to, the following:
 - a. Staging of Earthwork Activities. Scheduling the excavation and moving of soil materials to minimize the size of areas disturbed and unprotected from erosion for the shortest reasonable time.
 - b. Seeding. Seeding to protect disturbed areas occur as soon as reasonably possible following completion of that earthwork activity.
 - c. Mulching. Mulching to provide temporary protection of the soil surface from erosion.
 - d. Diversions. Diversions to divert water from work areas and to collect water from work areas for treatment and safe disposition. These are temporary and must be removed and the area restored to its near-original condition when the diversions are no longer required or when permanent measures are installed.
 - e. Stream Crossings. Culverts or bridges where equipment must cross streams. These are temporary must and be removed and the area restored to its near-original condition when the crossings are no longer required or when permanent measures are installed.
 - f. Sediment Basins. Sediment basins for collecting, settling, and eliminating sediment from eroding areas that impact properties and streams below the

construction sites. These basins are temporary and must be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.

- g. Sediment Filters. Straw bale filters or geotextile sediment fences for trapping sediment from areas with limited runoff. Sediment filters must be properly anchored to prevent erosion under or around them. Silt fences must be installed and maintained in accordance with ASTM D6462. These filters are temporary and must be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.
- h. Waterways. Waterways for the safe disposal of runoff from fields, diversions, and other structures or measures. These works are temporary and must be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.
- i. Other. Additional protection measures as specified in Section G of this specification or required by the federal, state, or local government.

C. Chemical Pollution

1. The contractor must provide watertight tanks or barrels or construct a sump sealed with plastic sheets to collect and temporarily contain chemical pollutants, such as drained lubricating or transmission fluids, grease, soaps, concrete mixer wash water, or asphalt, produced as a by-product from the construction activities. Dispose of pollutants in accordance with appropriate state and federal regulations. At the completion of the construction work, remove tanks, barrels, and sumps and restore the area to its original condition as specified in Section H of this specification. Sump removal must be conducted without causing pollution.
2. Sanitary facilities, such as chemical toilets, and septic tanks must not be located next to live streams, wells, or springs. They must be located at a distance sufficient to prevent contamination of any water source. At the completion of construction activities, facilities must be disposed of without causing pollution as specified in Section H of this specification.

D. Air Pollution

1. The burning of brush or slash and the disposal of other materials must adhere to state and local regulations.
2. Fire prevention measures must be taken to prevent the start or spreading of wildfires that may result from project activities. Firebreaks or guards must be constructed and maintained at locations shown on the drawings.
3. All public access or haul roads used by the contractor during construction of the project must be sprinkled or otherwise treated to fully suppress dust. All dust control methods must ensure safe construction operations at all times. If chemical dust suppressants are applied, the material must be a commercially available product specifically designed for dust suppression, and the application must follow the manufacturer's requirements and recommendations. A copy of the product data sheet and manufacturer's recommended application procedures must be provided to the engineer 5 working days before the first application.

E. Maintenance, Removal, and Restoration

1. All pollution-control measures and temporary works must be adequately maintained in a functional condition for the duration of the construction period. Remove all temporary measures and restore the site to near-original condition.

F. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, each item is measured to the nearest unit applicable. Payment for each item is made at the contract unit price for that item. For water or chemical suppressant items used for dust control for which items of work are established in Section H of this specification, measurement for payment will not include water or chemical suppressants that are used inappropriately or excessively. Such payment will constitute full compensation for the completion of the work.
2. Method 2. For items of work for which lump-sum prices are established in the contract, payment is made as the work proceeds and is supported by invoices presented by the contractor that reflect actual costs. If the total of all progress payments is less than the lump-sum contract price for this item, the balance remaining for this item will be included in the final contract payment. Payment of the lump-sum contract price will constitute full compensation for completion of the work.
3. Method 3. For items of work for which lump-sum prices are established in the contract, payment will be prorated and provided in equal amounts on each monthly progress payment estimate. The number of months used for prorating must be the number estimated to complete the work as outlined in the contractor's approved construction schedule. The final month's prorate amount will be provided with the final contract payment. Payment as described will constitute full compensation for completion of the work.
4. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items, and the items to which they are made subsidiary, are identified in Section H of this specification.

G. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Subsidiary Item, Pollution Control

- (1) The Contractor shall exercise precaution throughout the duration of the Contract to prevent the pollution of rivers and streams. Chemicals, fuels, lubricants, raw sewage, and other harmful wastes shall not be discharged into or alongside any river, stream, or natural or man-made waterway leading thereto. Materials identified as chemical pollutants in Section C of this specification shall be removed from the site and disposed of in compliance with applicable state and federal regulations. Any chemical pollutant spills shall be reported to the State Center for Pollution, Toxic Chemical and Oil Spills hotline at 1-(800)-642-3074.
- (2) The Contractor shall schedule clearing and grubbing operations in a manner which will minimize the amount of erodible surface soil exposed at any time. Areas shall not be cleared and grubbed until immediately prior to actual need or use of the area.
- (3) Pollution control measures shall be installed in a timely manner to protect against erosion from areas to be exposed by excavation and fill operations.
- (4) Equipment shall not be operated in live streams except as may be required to construct stream bank stabilization measures and temporary structures located in live streams. Frequent fording of live streams with construction equipment will not be permitted.

Temporary bridges, culverts, or other structures shall be installed wherever an appreciable number of stream crossings are necessary.

- (5) Protective cover shall be applied to disturbed areas as soon as work has been completed in the area, and/or whenever work in the disturbed area will be delayed or stopped more than five (5) continuous days. Areas to be provided with protective cover will be staked and/or flagged in the field by the Contracting Officer's Representative. Protective cover shall be applied to disturbed areas by mulching with straw. Materials and workmanship for providing protective cover shall be in accordance with Construction Specification 6, Seeding, Sprigging and Mulching.

Straw mulch shall be applied uniformly to disturbed areas at the rate of 2 tons per acre.

- (6) In Section F, Measurement and Payment, shall not apply. No separate payment shall be made for Pollution Control. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62 for Site 1 and Bid Item 5B, Gabions, Construction Specification 64 for Site 2.**

Construction Specification 6 — Seeding, Sprigging, and Mulching

Title 210 - Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of preparing the area for treatment; furnishing and placing seed, sprigs, mulch, fertilizer, inoculant, lime, and other soil amendments; and anchoring mulch in designated areas as specified.
2. The following BioPreferred® product categories are applicable to this specification:
 - a. Mulch and compost materials
 - b. Erosion control materials
 - c. Fertilizers
 - d. Agricultural spray adjuvants

B. Material

1. Seed.
 - a. All seed must conform to the current rules and regulations of the state where it is being used and must be from the latest crop available. It must meet or exceed the standard for purity and germination listed in Section G.
 - b. Seed must be labeled in accordance with the state laws and the U.S. Department of Agriculture rules and regulations under the Federal Seed Act in effect on the date of invitations for bids. Bag tag figures are evidence of purity and germination. No seed will be accepted with a test date of more than 9 months before the delivery date to the site.
 - c. Seed that has become wet, moldy, or otherwise damaged in transit or storage will not be accepted. The percent of noxious weed seed allowable must be as defined in the current state laws relating to agricultural seeds. Each type of seed must be delivered in separate sealed containers and fully tagged unless an exception is granted in writing by the contracting officer.
2. Fertilizer. Unless otherwise specified, the fertilizer must be a commercial-grade fertilizer. It must meet the standard for grade and quality specified by state law. Where fertilizer is furnished from bulk storage, the contractor must furnish a supplier's certification of analysis and weight. When required by the contract, a representative sample of the fertilizer must be furnished to the contracting officer for chemical analysis.

3. Inoculants. The inoculant for treating legume seeds must be a pure culture of nitrogen-fixing bacteria prepared specifically for the species and must not be used later than the date indicated on the container or as otherwise specified. A mixing medium, as recommended by the manufacturer, must be used to bond the inoculant to the seed. Two times the amount of the inoculant recommended by the manufacturer must be used (unless seed is applied using a hydraulic seeder; then four times the amount must be used). Seed must be sown within 24 hours of treatment and must not remain in the hydraulic seeder longer than 4 hours.
4. Lime and Other Soil Amendments. Lime must consist of standard ground agriculture limestone or approved equivalent. Standard ground agriculture limestone is defined as ground limestone meeting current requirements of the state's Department of Agriculture. Other soil amendments must meet quality criteria and application requirements specified in Section G.
5. Mulch Tackifiers. Asphalt emulsion tackifiers must conform to the requirements of ASTM D977, Specification for Emulsified Asphalt. The emulsified asphalt may be rapid setting, medium setting, or slow setting. Non-asphaltic tackifiers required because of environmental considerations must be as specified in Section G.
6. Straw Mulch Material. Straw mulch must consist of wheat, barley, oat or rye straw, hay, grass cut from native grasses, or other plants as specified in Section G. The mulch material must be air-dried reasonably light in color and must not be musty, moldy, caked, or otherwise of low quality. The use of mulch that contains noxious weeds is not permitted. The contractor must provide a method satisfactory to the contracting officer for determining weight of mulch furnished.
7. Other Mulch Materials. Mulching materials, such as wood cellulose fiber mulch, mulch tackifiers, synthetic fiber mulch, netting, and mesh may be required for specialized locations and conditions. These materials, when specified, must be accompanied by the manufacturer's recommendations for methods of application.

C. Seeding Mixtures, Sod, Sprigs, and Dates of Planting

1. The application rate per acre for seed mixtures, sprigs, or sod and the date of seeding or planting must be as shown on the plans or as specified in Section G.

D. Seedbed Preparation and Treatment

1. Dress areas to be treated to a smooth, firm surface. On sites where equipment can operate on slopes safely, adequately loosen and smooth the seedbed (4 to 6 inches deep). Depending on soil and moisture conditions, disking, cultipacking, or both, may be necessary to properly prepare a seedbed. Where equipment cannot operate safely, prepare the seedbed by methods like scarifying, providing a roughened soil surface by hand so that broadcast seed will remain in place.
2. If seeding is to be accomplished immediately following construction operations, seedbed preparation may not be required except on a compacted, polished, or freshly cut soil surface.

3. Rocks larger than 6 inches in diameter, trash, weeds, and other debris that will interfere with seeding or maintenance operations must be removed or disposed of as specified in Section G.
4. Discontinue seedbed preparation when soil moisture conditions are not suitable for the preparation of a satisfactory seedbed as determined by the responsible engineer.

E. Seeding, Sprigging, Fertilizing, Mulching, and Stabilizing

1. Perform all seeding or sprigging operations in such a manner that the seed or sprigs are applied in the specified quantities uniformly in the designated areas. The method and rate of seed application must be as specified in Section G. Unless otherwise specified, seeding or sprigging must be accomplished within 2 days after final grading is completed and approved.
2. Apply fertilizer, lime, and other soil amendments as specified in Section G. When specified, the fertilizer and soil amendments must be thoroughly incorporated into the soil immediately following surface application.
3. The rate, amount, and kind of mulching or mesh must be as specified in Section G. Apply mulches uniformly to the designated areas. They must be applied to areas seeded not later than 2 working days after seeding has been performed. Stabilize straw mulch material within 24 hours of application using a mulch crimper or equivalent anchoring tool or by a suitable tackifier. When the mulch crimper or equivalent anchoring tool is used, it must have straight blades and be the type manufactured expressly for and capable of firmly punching the mulch into the soil. Where the equipment can be safely operated, it must be operated on the contour. Use hand methods where equipment cannot safely operate to perform the work required.
4. Apply the tackifier uniformly over the mulch material at the specified rate or inject it into the mulch material as it is being applied. Stabilizing materials like mesh or netting must be applied smoothly but loosely on the designated areas. The edges of these materials must be buried or securely anchored using spikes or staples as specified in Section G.
5. The contractor must maintain the mesh or netting areas until all work under the contract has been completed and accepted. Maintenance must consist of the repair of areas damaged by water erosion, wind, fire, or other causes. Such areas must be repaired to reestablish the intended condition and to the design lines and grades required by the contract. The areas must be re-fertilized, reseeded, and re-mulched before the new application of the mesh or netting.

F. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, each area treated is measured as specified in Section G and area is calculated to the nearest 0.1 acre. Payment for treatment is made at the contract unit price for the designated treatment, which will constitute full compensation for completion of the work.

- a. When specified as an item of work, mesh or netting is measured to the nearest square yard of surface area covered and accepted. Payment is made at the contract unit price and will constitute full compensation for completion of the work.
2. Method 2. For items of work for which specific lump-sum prices are established in the contract, the quantity of work will not be measured for payment. Payment for this item is made at the contract lump-sum price for the item and will constitute full compensation for the completion of the work.
3. Method 3. For items of work for which lump-sum prices are established in the contract, payment is made as the work proceeds. Progress payments will be determined as specified in Section G. Payment of the lump-sum contract price will constitute full compensation for completion of the work.
4. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section G.

G. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. General

- (1) This item shall consist of seeding and mulching to permanent cover all disturbed areas due to work associated with the construction activities.
- (2) No permanent seeding shall be done between October 1 and December 1 unless otherwise approved by the Contracting Officer.
- (3) The approved lines and grades for permanent seeding at the time of permanent seeding shall be maintained during and after the seeding and mulching operation. Any area that has been damaged or had part or all of the seeding material removed by any means shall be repaired before acceptance will be made. The area to be repaired shall be filled and/or graded as needed to bring it back to neat lines, fertilized, seedbed prepared, and then seeded and mulched at the specified rates.
- (4) Lime shall be applied at an application rate of 2 tons per acre.
- (5) The following minimum amounts of fertilizer shall be applied per acre prior to seeding:
 - a. Nitrogen (N) 40 lbs.
Phosphoric Acid (P₂O₅) 80 lbs.
Potash (K₂O) 80 lbs.
 - b. The required application can be achieved by applying fertilizer having an analysis of 10-20-20 at the rate of 400 pounds per acre or 9 pounds per 1000 square feet.
- (6) Seedbed Preparation:
 - a. Lime and fertilizer shall be applied prior to final seedbed preparation.
 - b. Seeding shall be accomplished immediately after achieving final grade. Seedbed preparation shall only be required on compacted, polished, or freshly cut soil surface. Where special seedbed preparation is required, the hardened soil shall be loosened to a depth of 1 ½ inches; either by equipment or manual methods.
- (7) Seed:
 - a. Each seed species shall comply with the West Virginia Department of Agriculture's standards for purity and germination. Unless otherwise directed by the Contracting Officer's Representative, the seed mixtures and application rates are:

<u>Seed</u>	<u>Rate, lb/ac.</u>
Bluegrass	8
Kentucky 31 Tall Fescue	10
Annual Ryegrass	15
Red Fescue	5
Perennial Ryegrass	5

- b. When seeding is performed between April 15 – August 1 or December 1 – March 1, the specified rates of seed per acre shall be increased by 50 percent.
- (8) Mulch material shall be wheat, oat or rye straw only. Mulch shall be applied uniformly over all seeded areas at the specified rate per acre. Application shall be within 24 hours after seeding. A mechanical blower may be used to apply the mulch. Application rates and material requirements shall be in accordance with the following table:

**Time Period vs. Mulching Requirements for
Revegetation of Construction Sites**

<u>Time Period</u>	<u>Suitability</u>	<u>Mulch Material</u>	<u>Rate/Acre</u>
March 1-April 15 or August 1-October 1	Best seeding periods	Small grain straw ¹	2 tons
April 15-August 1	High Risk Period	Small grain straw ¹	2 ½ tons
October 1-Dec. 1	Do not seed	Small grain straw ¹	2 tons
Dec. 1-March 1	Good seeding period.	Small grain straw ¹	2 ½ tons

¹ Small grain straw shall consist of wheat, oat, or rye straw.

Erosion control matting shall be used in place of straw mulch in areas 3H:1V or steeper. Erosion control matting shall be a commercially prepared composite turf reinforcement mat containing a 70 percent straw and 30 percent coconut fiber matrix incorporated into an extended temporary control mat with biodegradable netting, minimum weight of 10.5 oz/square yard, rated for 2:1 or steeper slopes, and 2.02 lbs./sf unvegetated shear stress.

b. Bid Item 1A, Seeding and Mulching, Site 1

- (1) This item shall consist of seeding and mulching to permanent cover all disturbed areas due to work associated with the construction activities at Site 1.
- (2) In Section F, Measurement and Payment, Method 1 shall apply.

c. Bid Item 1B, Seeding and Mulching, Site 2

- (2) This item shall consist of seeding and mulching to permanent cover all disturbed areas due to work associated with the construction activities at Site 2.
- (2) In Section F, Measurement and Payment, Method 1 shall apply.

Construction Specification 8 — Mobilization and Demobilization

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the mobilization and demobilization of the contractor's forces and equipment necessary for performing the work required under the contract. It does not include mobilization and demobilization for specific items of work for which payment is provided elsewhere in the contract. Mobilization will not be considered as work in fulfilling the contract requirements for commencement of work.

B. Equipment and Material

1. Mobilization includes all activities and associated costs for transportation of contractor's personnel, equipment, and operating supplies to the site; establishment of offices, buildings, and other necessary general facilities for the contractor's operations at the site; premiums paid for performance and payment bonds including coinsurance and reinsurance agreements as applicable; and other items specified in Section D of this specification.
2. Demobilization includes all activities and costs for transportation of personnel, equipment, and supplies not required or included in the contract from the site, including the disassembly, removal, and site cleanup of offices, buildings, and other facilities assembled on the site specifically for this contract.
3. This work includes mobilization and demobilization required by the contract at the time of award. If additional mobilization and demobilization activities and costs are required during the performance of the contract as a result of changed, deleted, or added items of work for which the contractor is entitled to an adjustment in contract price, compensation for such costs will be included in the price adjustment for the item or items of work changed or added.

C. Payment

1. Payment will be made as the work proceeds after presentation of paid invoices or documentation of direct costs by the contractor showing specific mobilization and demobilization costs and supporting evidence of the charges of suppliers, subcontractors, and others. When the total of such payments is less than the lump-sum contract price, the balance remaining will be included in the final contract payment. Payment of the lump sum contract price for mobilization and demobilization will constitute full compensation for completion of the work.

2. Payment will not be made under this item for the purchase costs of materials having a residual value, the purchase costs of materials to be incorporated in the project, or the purchase costs of operating supplies.

D. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. General, Mobilization and Demobilization

- (1) This item shall consist of mobilization and demobilization of all work forces, materials, and equipment necessary to complete the work.
- (2) The contractor will be required to provide a portable toilet at the sites.
- (3) The Contractor shall contact MISS Utility before work at the site begins. The Contractor shall provide the MISS Utility confirmation number for the site to the Contracting Officer's Representative before commencement of the work.
- (4) Prior to construction the Contractor shall verify that all utilities have been disconnected at the described sites.
- (5) The Contractor may use a portable sign for public notice and for posting documents as required by the permits and this contract. Signage may be prepared to accommodate frequent revisions for use at multiple sites.

b. Bid Item 2A, Mobilization and Demobilization, Site 1

- (1) This item shall consist of mobilization and demobilization of all work forces, materials, and equipment necessary to complete the work at Site 1.
- (2) In Section C, Payment shall apply upon approval of completed work by the Contracting Officer's Representative.

c. Bid Item 2B, Mobilization and Demobilization, Site 2

- (1) This item shall consist of mobilization and demobilization of all work forces, materials, and equipment necessary to complete the work at Site 2.
- (2) In Section C, Payment shall apply upon approval of completed work by the Contracting Officer's Representative.

Construction Specification 9 — Traffic Control

Title 210 – Engineering

Part 642 – Specifications

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of establishing traffic control and maintaining safe, convenient use of public roads and right of way.

B. Traffic and Access

1. The contractor's operations must cause no unnecessary inconvenience to the public. Maintain the public right of way at all times unless interruption is authorized by proper local authority. Contractor's authorized closing or detour plans must be provided to the engineer for approval.
2. Provide and maintain safe and adequate access to all public protection devices and to all critical utility-control locations. Facility access must be continuous and unobstructed unless otherwise approved.

C. Storage of Equipment and Material in Public Streets

1. Construction materials and equipment must not be stored or parked on public streets, roads, or highways. During any material or equipment loading or unloading activities that may temporarily interfere with traffic, an acceptable detour must be provided for the duration of the activity. Any associated expense for this activity is the responsibility of the contractor.
2. Excavated material, including suitable material that is intended for adjacent trench backfill or other earth backfill as specified in Section E of this specification, must not be stored on public streets, roads, or highways that remain in service for the public. Any waiver of this requirement must be obtained from the proper local authority and approved by the engineer. All excess and unsuitable material must be removed from the site as soon as possible. Remove any spillage from roadways before they are used by the public.

D. Street Closures, Detours, and Barricades

1. The contractor must comply with the requirements of all applicable, responsible units of government for closure of any street, road, or highway. The contractor must provide the required barriers, guards, lights, signs, temporary bridges, and flaggers and inform the public of any detours and construction hazards by the most suitable means available, such as local newspapers or radio stations. The contractor is also responsible for compliance with additional public safety requirements that may arise during construction. The contractor must furnish,

install, and, upon completion of the work, promptly remove all signs, warning devices, and other materials used in the performance of this work.

2. Unless otherwise specified, the contractor must notify, in writing, the fire chief, police chief, county sheriff, state patrol, schools that operate school buses, or any other government official as appropriate no less than 7 days before closing, partly closing, or reopening any street, road, or highway.
3. Unless otherwise specified, the contractor must furnish to the engineer a written plan showing the proposed method of signing, barricading for traffic control, and safety for street detours and closures.
4. All temporary detours will be maintained to ensure use of public rights of way is provided in a safe manner. This may include dust control, grading, and graveling as required in Section G of this specification.

E. General and Specific References

1. All signs, signals, barricades, use of flaggers, and other traffic control and public safety devices must conform to the general requirements set forth in the Manual of Uniform Traffic Control Devices (MUTCD) and the latest edition of Standard Highway Signs and Standard Alphabets for Highway Signs and OSHA Construction Industry Standards (29 CFR Part 1926), Subpart G, Signs, Signals, and Barricades, unless otherwise specified in Section G of this specification.

F. Measurement and Payment

1. For items of work for which specific lump-sum prices are established in the contract, payment for the work is made at the contract lump-sum price. Progress payments will be made based upon the percentage of estimated total time that traffic control will be required unless otherwise specified in Section G of this specification. Payment will constitute full compensation for all flaggers, labor, materials, equipment, and other items necessary and incidental to completion of the work.
2. Compensation for any item of work described in the contract but not listed in the bid schedule will be included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in Section G of this specification.

G. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Subsidiary Item, Traffic Control

- (1) Temporary blockages of the road are permitted with proper flagging provisions. A minimum of one lane of traffic shall be maintained at all times, with two lane traffic restored after work hours.
- (2) Roadways will need to be kept clean of dirt, debris, and any other obstructions to allow for the safe passage of vehicles.
- (3) In Section F, Measurement and Payment, shall not apply. No separate payment shall be made for Traffic Control. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 11 — Removal of Water

Tile 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the removal of surface water and ground water as necessary to perform the construction required by the contract in accordance with the specifications. It must include: (1) constructing, installing, building, and maintaining all necessary temporary water containment facilities, channels, and diversions; (2) furnishing, installing, and operating all necessary pumps, piping, and other facilities and equipment; and (3) removing all such temporary works and equipment after their intended function is no longer required.

B. Diverting Surface Water

1. The contractor must install, maintain, and operate all cofferdams, channels, flumes, sumps, and all other temporary diversion and protective works needed to divert streamflow and other surface water through or around the construction site. Control of surface water must be continuous during the period that damage to construction work could occur. Unless otherwise specified or approved, the diversion outlet must be into the same drainageway that the water would have reached before being diverted.
2. The contractor must furnish the contracting officer, in writing, a proposed plan for diverting surface water before beginning any construction activities for which a diversion is required, unless waived in Section H of this specification. Acceptance of this plan or the waiving of the plan requirement will not relieve the contractor of the responsibilities related to this activity during the process of completing the work as specified.

C. Dewatering the Construction Site

1. Foundations, cutoff trenches, and all other parts of the construction site must be dewatered and kept free of standing water and muddy conditions as necessary for the proper execution of the work. The contractor must furnish, install, operate, and maintain all drains, sumps, pumps, casings, well points, and all other equipment required to properly dewater the site as specified. Dewatering systems that cause a loss of soil fines from the foundation areas will not be permitted.
2. The contractor must furnish the contracting officer, in writing, a proposed plan for dewatering before commencing with any construction activity for which dewatering may be required unless waived in Section H of this specification.

Acceptance of this plan or the waiving of the plan requirement will not relieve the contractor of the responsibilities for completing the specified work.

D. Dewatering Borrow Areas

1. The contractor must maintain all borrow areas free of surface water or otherwise provide for timely and effective removal of surface and subsurface water that accumulates within the borrow area unless waived in Section H of this specification. Process borrow material as necessary to achieve proper and uniform moisture content at the time of placement.
2. If pumping to dewater borrow areas is included as a bid item of work in the bid schedule, equip each pump discharge pipe with a water meter. The meter must be such that the measured quantity of water is accurate within 3 percent of the true quantity. The contractor must provide necessary support to perform accuracy tests of the water meter when requested by the contracting officer.

E. Erosion and Pollution Control

1. Removal of water from the construction site, including the borrow areas, must be accomplished so that erosion and the transporting of sediment and other pollutants are minimized. Dewatering activities must be accomplished in a manner that the water table water quality is not altered. Pollution control activities must not conflict with the requirements of Construction Specification 5, Pollution Control, if it is a part of this contract.

F. Removal of Temporary Works

1. When temporary works are no longer needed, the contractor must remove and return the area to a condition similar to that which existed before construction. Areas where temporary works were located must be graded for sightly appearance and proper functioning and access to the installed works of improvement and may not obstruct natural surface waterflows. The contractor must exercise extreme care during the removal stages to minimize the loss of soil sediment and debris that was trapped during construction.
2. Pipes, casings, and any other material used to dewater the site must be removed from temporary wells. Fill the wells to ground level with clean gravel or other suitable material approved by the contracting officer. The contractor must exercise extreme care to prevent pollution of the ground water by these actions.

G. Measurement and Payment

1. Method 1.
 - a. Items of work listed in the bid schedule for removal of water, diverting surface water, and dewatering construction sites and borrow areas are paid for at the contract lump-sum prices. Such payment will constitute full compensation for all labor, equipment, tools, and all other items necessary and incidental to the completion of the work.
2. Method 2.

- a. Items of work listed in the bid schedule for removal of water, diverting surface water, dewatering construction sites, and dewatering borrow areas are paid for at the contract lump-sum prices. Such payment will constitute full compensation for furnishing, installing, operating, and maintaining the necessary trenches, drains, sumps, pumps, piping, and for all labor, equipment, tools, and all other items necessary and incidental to the completion of the work. The exception is that additional payment for pumping to dewater borrow areas and the removal of water will be made as described in the following paragraph.
 - b. If pumping to dewater borrow areas is a contract bid item, payment is made at the contract unit price, which must be the price per 1,000 gallons shown in the bid schedule. Such payment will constitute full compensation for pumping only. Compensation for equipment, preparation, and other costs associated with pumping is included in the lump-sum payment for removal of water or the lump-sum payment for dewatering the borrow areas. Payment is made only for pumping that is necessary to dewater borrow areas that cannot be effectively drained by gravity or that must have the water table lowered to be usable as a suitable borrow source. Pumping for other purposes will not be included for payment under this item.
3. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the contract line item to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section H of this specification.

H. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Subsidiary Item, Removal of Water

- (1) This item shall include all costs that are required for providing all materials, equipment, labor, and services necessary to divert surface waters and dewater the work areas as required to perform the work during the contract.
- (2) Submittal of written plans for diverting surface waters and dewatering the site are not required.
- (3) In Section G, Measurement and Payment, shall not apply. No separate payment shall be made for Removal of Water. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Grouted Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 21 — Excavation

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the excavation required by the drawings and specifications, as well as the disposal of the excavated materials.

B. Classification

1. Excavation is classified as common excavation, rock excavation, or unclassified excavation in accordance with the following definitions.
2. Common excavation is defined as the excavation of all materials that (1) can be excavated, transported, and unloaded using heavy ripping equipment and wheel tractor-scrapers with pusher tractors or (2) can be excavated and dumped into place or loaded onto hauling equipment by excavators having a rated capacity of 1 cubic yard or larger and equipped with attachments (shovel, bucket, backhoe, dragline, or clam shell) appropriate to the material type, character, and nature of the materials.
3. Rock excavation is defined as the excavation of all hard, compacted, or cemented materials that require blasting or the use of ripping and excavating equipment larger than defined for common excavation. The excavation and removal of isolated boulders or rock fragments larger than 1 cubic yard encountered in materials otherwise conforming to the definition of common excavation must be classified as rock excavation. The presence of isolated boulders or rock fragments larger than 1 cubic yard is not in itself sufficient cause to change the classification of the surrounding material.
 - a. For the purpose of these classifications, the following definitions apply:
 - (1) Heavy ripping equipment is a rear-mounted, heavy-duty, single-tooth, ripping attachment mounted on a track-type tractor having a power rating of at least 250-flywheel horsepower, unless otherwise specified in Section J.
 - (2) A wheel tractor-scraper is a self-loading (not elevating) and unloading scraper having a struck-bowl capacity of at least 12 cubic yards.
 - (3) A pusher tractor is a track-type tractor having a power rating of at least 250-flywheel horsepower equipped with appropriate attachments.

4. Unclassified excavation is defined as the excavation of all materials encountered, including rock materials, regardless of their nature or the manner in which they are removed.

C. Blasting

1. The transportation, handling, storage, and use of dynamite and other explosives must be directed and supervised by someone of proven experience and ability who is authorized and qualified to conduct blasting operations.
2. Blasting must be done in a manner that prevents damage to the work or unnecessary fracturing of the underlying rock materials. It must conform to any special requirements in Section J of this specification. When specified in Section J, the contractor must furnish the engineer, in writing, a blasting plan before blasting operations begin.

D. Use of Excavated Material

1. Method 1. To the extent they are needed, all suitable material from the specified excavations must be used in the construction of required permanent earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer. The contractor must not waste or otherwise dispose of suitable excavated material.
2. Method 2. Suitable material from the specified excavations may be used in the construction of required earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer.

E. Disposal of Waste Materials

1. Method 1. All surplus or unsuitable excavated materials are designated as waste and must be disposed of at the locations shown on the drawings.
2. Method 2. All surplus or unsuitable excavated materials are designated as waste and disposed of by the contractor at chosen sites away from the site of the work. The disposal must be in an environmentally acceptable manner that does not violate local rules and regulations.

F. Excavation Limits

3. Excavations must comply with the Occupational Safety and Health Administration Construction Industry Standards Subpart P, Excavations, Trenching, and Shoring (29 CFR pt. 1926). All excavations must be completed and maintained in a safe and stable condition throughout the total construction phase. Structure and trench excavations must be completed to the specified elevations and to the length and width required to safely install, adjust, and remove any forms, bracing, or supports necessary for the installation of the work. Excavations outside the lines and limits shown on the drawings or specified herein are required to meet safety requirements and must be the responsibility of the contractor in constructing and maintaining a safe, stable excavation.

G. Borrow Excavation

1. When the quantities of suitable material obtained from specified excavations are insufficient to construct the specified earthfills and earth backfills, obtain additional material from the designated borrow areas. The extent and depth of borrow pits within the limits of the designated borrow areas are as specified in Section J or as approved by the engineer.
2. Borrow pits must be excavated and finally dressed to blend with the existing topography. They are sloped to prevent ponding and to provide drainage.

H. Over Excavation

1. Correct excavation in rock beyond the specified lines and grades by filling the resulting voids with Portland Cement concrete. Materials and mix proportions must be approved by the engineer. Concrete that will be exposed to the atmosphere when construction is completed must meet the requirements of concrete selected for use under Construction Specification 31, Concrete for Major Structures, or 32, Structure Concrete, as appropriate.
2. Concrete that will be permanently covered must contain not less than five bags of cement per cubic yard. The concrete must be placed and cured as specified by the engineer.
3. Correct excavation in earth beyond the specified lines and grades by filling the resulting voids with approved compacted earthfill. The exception to this is that if the earth will become the subgrade for riprap, rockfill, sand or gravel bedding, or drainfill, the voids may be filled with material conforming to the specifications for the riprap, rockfill, bedding, or drainfill. Before correcting an over excavation condition, the contractor must review the planned corrective action with the engineer and obtain approval of the corrective measures.

I. Measurement and Payment

1. Unit Price Methods. For items of work for which specific unit prices are established in the contract, the volume of each type and class of excavation within the specified pay limits is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas or by methods outlined in Section J of this specification. Regardless of quantities excavated, the measurement for payment is made to the specified pay limits except when excavation outside the specified lines and grades directed by the engineer to remove unsuitable material is included. Excavation required because unsuitable conditions result from the contractor's improper construction operations, as determined by the engineer, is not included for measurement and payment.
2. Method 1. The pay limits are as designated on the drawings.
3. Method 2. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.

- b. The lower and lateral limits are the neat lines and grades shown on the drawings.
- 4. Method 3. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.
 - b. The lower and lateral limits is the true surface of the completed excavation as directed by the engineer.
- 5. Method 4. The pay limits are defined as follows:
 - a. The upper limit is the original ground surface as it existed before the start of construction operations, unless excavation is performed within areas designated for previous excavation or earthfill; then the upper limit is the modified ground surface resulting from the specified previous excavation or earthfill.
 - b. The lower limit is at the bottom surface of the proposed structure.
 - c. The lateral limits are 18 inches outside of the outside surface of the proposed structure or are the vertical planes 18 inches outside of and parallel to the footings, whichever gives the larger pay quantity, except as provided in paragraph d below.
 - d. For trapezoidal channel linings or similar structures that are to be supported by the sides of the excavation without intervening forms, the lateral limits are at the underside of the proposed lining or structure.
 - e. For the purposes of the definitions in paragraphs b, c, and d above, any specified bedding or drainfill directly beneath or beside the structure will be considered to be a part of the structure.
- 6. All Methods. The following provisions apply to all methods of measurement and payment.
 - a. Payment for each type and class of excavation is made at the contract unit price for that type and class of excavation. Such payment will constitute full compensation for all labor, materials, equipment, and other items necessary and incidental to the performance of the work with the exception that extra payment for backfilling over excavation will be made in accordance with the following provisions.
 - b. Payment for backfilling over excavation, as specified in Section H of this specification, is made only if the excavation outside specified lines and grades is directed by the engineer to remove unsuitable material and the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.
 - c. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section J of this specification.

J. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Excavation, General Requirements

- 1) This item shall consist of all excavation necessary to perform the work, as shown on the drawings and staked in the field.
- 2) All excavation shall be unclassified.
- 3) The contractor shall use care to stockpile any topsoil material to be used in the reclamation of the site.
- 4) No blasting shall be permitted.
- 5) The excavated material shall be used in accordance with Construction Specification 23, Earthfill and Construction Specification 25, Rockfill. Any excess material shall be disposed of in accordance with Construction Specification 4, Channel Clearing and Shaping.

b. Subsidiary Item, Excavation

- 1) This item shall consist of providing any and all materials, equipment, and labor necessary to excavate material necessary to complete the work.
- 2) In Section I, Measurement and Payment, shall not apply. No separate payment shall be made for Excavation. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 23 — Earthfill

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the construction of earth embankments, other earthfills, and earth backfills required by the drawings and specifications.
2. Earthfill is composed of natural earth materials that can be placed and compacted by construction equipment operated conventionally.
3. Earth backfill is composed of natural earth material placed and compacted in confined spaces or adjacent to structures (including pipes) by hand tamping, manually directed power tampers or vibrating plates, or their equivalent.

B. Material

1. All fill material must be obtained from required excavations and designated borrow areas. The selection, blending, routing, and disposition of material in the various fills must be subject to approval by the engineer.
2. Fill materials must contain no frozen soil, sod, brush, roots, or other perishable material. Rock particles larger than the maximum size specified for each type of fill must be removed prior to compaction of the fill.
3. The types of material used in the various fills must be as listed and described in the specifications and drawings.

C. Foundation preparation

1. Foundations for earthfill must be stripped to remove vegetation and other unsuitable material or must be excavated as specified.
2. Except as otherwise specified, earth foundation surfaces must be graded to remove surface irregularities and must be scarified parallel to the axis of the fill or otherwise acceptably scored and loosened to a minimum depth of 2 inches. The moisture content of the loosened material must be controlled as specified for the earthfill and the surface material of the foundation must be compacted and bonded with the first layer of earthfill as specified for subsequent layers of earthfill.
3. Earth abutment surfaces must be free of loose, uncompacted earth in excess of 2 inches in depth normal to the slope and must be at such a moisture content that the earthfill can be compacted against them to produce a good bond between the fill and the abutments.
4. Rock foundation and abutment surfaces must be cleared of all loose material by hand or other effective means and must be free of standing water when fill is

placed upon them. Occasional rock outcrops in earth foundations for earthfill, except in dams and other structures designed to restrain the movement of water, must not require special treatment if they do not interfere with compaction of the foundation and initial layers of the fill or the bond between the foundation and the fill.

5. Foundation and abutment surfaces must be no steeper than one horizontal to one vertical unless otherwise specified. Test pits or other cavities must be filled with compacted earthfill conforming to the specifications for the earthfill to be placed upon the foundation.

D. Placement

1. Earthfill must not be placed until the required excavation and foundation preparation have been completed and the foundation has been inspected and approved by the engineer. Earthfill must not be placed upon a frozen surface nor have snow, ice, or frozen material incorporated in the earthfill matrix.
2. Earthfill must be placed in approximately horizontal layers. The thickness of each layer before compaction must not exceed the maximum thickness specified in section J or shown on the drawings. Materials placed by dumping in piles or windrows must be spread uniformly without exceeding the specified thickness before being compacted.
3. Hand-compacted earth backfill must be placed in layers whose thickness before compaction does not exceed the maximum thickness specified for layers of earth backfill compacted by manually directed power tampers.
4. Place earth backfill in a manner that prevents damage to the structures and allows the structures to assume the loads from the earth backfill gradually and uniformly. The height of the earth backfill adjacent to a structure must be increased at approximately the same rate on all sides of the structure.
5. Meet the following additional requirements for earthfill and earth backfill in dams, levees, and other structures designed to restrain the movement of water.
 - a. The distribution of materials throughout each zone must be essentially uniform and the earthfill free from lenses, pockets, streaks, or layers of material differing substantially in texture, moisture content, or gradation from the surrounding material. Zone earthfills must be constructed concurrently unless otherwise specified.
 - b. Scarified the surface of each layer parallel to the axis of the fill to at minimum a depth of 2 inches before the next layer is placed.
 - c. The top surface of embankments must be maintained approximately level during construction with two exceptions: a crown or cross-slope of about 2 percent must be maintained to ensure effective drainage, or as otherwise specified for drainfill or sectional zones.
 - d. Construct dam embankments in continuous layers from abutment to abutment except where openings to facilitate construction or to allow the passage of streamflow during construction are specifically authorized in the contract.

- e. Embankments built at different levels as described under paragraphs c or d above must be constructed so that the slope of the bonding surfaces between embankments in place and embankments to be placed is not steeper than 3 feet horizontal to 1 foot vertical. The bonding surface of the embankment in place must be stripped of all material not meeting the requirements of this specification and must be scarified, moistened, and recompactd when the new earthfill is placed against it. This ensures a good bond with the new earthfill and obtains the specified moisture content and density at the contact of the in-place and new earthfills.

E. Control of moisture content

1. During placement and compaction of earthfill and earth backfill, the moisture content of the material being placed must be maintained within the specified range.
2. The application of water to the earthfill material must be accomplished at the borrow areas insofar as practicable. Water may be applied by sprinkling the material after placement on the earthfill, if necessary. Obtain a uniform moisture distribution by disking.
3. Material that is too wet when deposited on the earthfill must either be removed or dried to the specified moisture content prior to compaction.
4. If the top surface of the preceding layer of compacted earthfill or a foundation or abutment surface in the contact zone with the earthfill becomes too dry to permit suitable bond, it must either be removed or scarified and moistened by sprinkling to an acceptable moisture content before placement of the next layer of earthfill.

F. Compaction

1. Earthfill. Earthfill must be compacted according to the following requirements for the class of compaction specified:
 - a. Class A compaction. Compact each layer of earthfill as necessary to provide the density of the earthfill matrix not less than the minimum density specified in Section J or identified on the drawings. The earthfill matrix is defined as the portion of the earthfill material finer than the maximum particle size allowed in the reference compaction test method specified (ASTM D698 or ASTM D1557).
 - b. Class B compaction. Compact each layer of earthfill to a mass density not less than the minimum density specified.
 - c. Class C compaction. Compact each layer of earthfill by the specified number of passes of the type and weight of roller or other equipment specified or by an equivalent approved method. Each pass must consist of at least one passage of the roller wheel or drum over the entire surface of the layer.
2. Earth backfill. Compact earth backfill adjacent to structures to a density equivalent to that of the surrounding in-place earth material or adjacent required earthfill or earth backfill. Compaction must be accomplished by hand tamping or manually directed power tampers, plate vibrators, walk-behind, miniature, or self-

propelled rollers. Unless otherwise specified, heavy equipment, including backhoe-mounted power tampers or vibrating compactors, and manually directed vibrating rollers must not be operated within 3 feet of any structure. Towed or self-propelled vibrating rollers must not be operated within 5 feet of any structure. Compaction by means of drop weights operating from a crane or hoist is not permitted.

3. Heavy Equipment. The passage of heavy equipment will not be allowed:
 - a. Over cast-in-place conduits within 14 days after placement of the concrete
 - b. Over cradled or bedded precast conduits within 7 days after placement of the concrete cradle or bedding
 - c. Over any type of conduit until the backfill has been placed above the top surface of the structure to a height equal to one-half the clear span width of the structure or pipe or 3 feet, whichever is greater, except as may be specified in Section J.
4. Compacting adjacent to structures. Compacting of earth backfill adjacent to structures must not be started until the concrete has attained the strength specified in Section J for this purpose. The strength is determined by compression testing of test cylinders cast by the contractor's quality control personnel for this purpose and cured at the worksite in the manner specified in ASTM C31 for determining when a structure may be put into service.
5. Placement Interval Against Concrete. When the required strength of the concrete is not specified as described above, compaction of earth backfill adjacent to structures must not be started until the following time intervals shown in figure 23-1 have elapsed after placement of the concrete.

Figure 23-1 Time interval before placing backfill

Structure	Time interval (days)
Vertical or near-vertical walls with earth loading on one side only	14
Walls backfilled on both sides simultaneously	7
Conduits and spillway risers, cast-in-place (with inside forms in place)	7
Conduits and spillway risers, cast-in-place (inside forms removed)	14
Conduits, pre-cast, cradled	2
Conduits, pre-cast, bedded	1
Cantilever outlet bents (backfilled both sides simultaneously)	3

G. Reworking or removal and replacement of defective earthfill

1. Earthfill placed at densities lower than the specified minimum density or at moisture contents outside the specified acceptable range of moisture content, or otherwise not conforming to the requirements of the specifications must be reworked to meet the requirements or removed and replaced by acceptable earthfill. The replacement earthfill and the foundation, abutment, and earthfill surfaces upon which it is placed must conform to all requirements of this

specification for foundation preparation, approval, placement, moisture control, and compaction.

H. Testing

1. During the course of the work, the contractor must perform quality control tests, as applicable, to identify earthfill and earth backfill materials; determine the reference maximum density and optimum moisture content; and document that the moisture content of material at the time of compaction and the density of earthfill and earth backfill in place conform to the requirements of this specification.
2. Determining Reference Maximum Density and Optimum Moisture Content. For Class A compaction, the reference maximum density and optimum moisture content must be determined in accordance with the compaction test and method specified on the drawings or in section J.
3. Documenting Specification Conformance. In-place densities of earthfill and earth backfill requiring Class A compaction must be measured in accordance with ASTM D1556, D2167, D2937, D6938, or D8167. Moisture contents of earthfill and earth backfill at the time of compaction must be measured in accordance with ASTM D2216, D4643, or D6938. Values of moisture content determined by ASTM D2216 are considered the true value of the soil moisture. Values of moisture content determined by ASTM D4643 or D6938 must be verified by comparison to values obtained by ASTM D2216. Values of in-place density and moisture content determined by these tests must be compared to the minimum density and moisture content range specified on the drawings or in Section J.
4. Correction for Oversize Particles. If the materials to be used for earthfill or earth backfill contain more than 5 percent by dry weight of oversize rock particles (particles larger than those allowed in the specified compaction test and method), corrections for oversize particles must be made using the appropriate procedures explained in ASTM D4718.

I. Measurement and payment

1. Unit Price Methods. For items of work for which specific unit prices are established in the contract, the volume of each type and compaction class of earthfill and earth backfill within the specified zone boundaries and pay limits is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas. Unless otherwise specified in Section J, no deduction in volume is made for embedded items such as, but not limited to, conduits, inlet structures, outlet structures, embankment drains, sand diaphragm and outlet, and their appurtenances.
2. Pay Limits. The pay limits must be as defined below, with the further provision that earthfill required to fill voids resulting from over excavation of the foundation, outside the specified lines and grades, will be included in the measurement for payment only under the following conditions:

- a. Where such over excavation to remove unsuitable material is directed by the engineer
 - b. Where the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.
3. Beyond Specified Lines and Grades. Earthfill beyond the specified lines and grades to backfill excavation required for compliance with OSHA requirements will be considered subsidiary to the earthfill bid items.
4. Method 1. The pay limits must be as designated on the drawings.
5. Method 2. The pay limits must be the measured surface of the foundation when approved for placement of the earthfill and the specified neat lines of the earthfill surface.
6. Method 3. The pay limits must be the measured surface of the foundation when approved for placement of the earthfill and the measured surface of the completed earthfill.
7. Method 4. The pay limits must be the specified pay limits for excavation and the specified neat lines of the earthfill surface.
8. Method 5. The pay limits must be the specified pay limits for excavation and the measured surface of the completed earthfill.
9. Method 6. Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and other items necessary and incidental to the performance of the work.
10. Method 7. Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and other items necessary and incidental to the performance of the work except furnishing, transporting, and applying water to the foundation and earthfill material. Water applied to the foundation and earthfill material is measured and payment made as specified in Construction Specification 10.
11. All methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section J of this specification.

J. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Earthfill, General Requirements

- 1) No foundation preparation is required beyond removing debris from within the designated fill area.
- 2) No testing of placed earthfill is required.
- 3) Excavated deposition material that will be used for Earthfill shall be approved by the Contracting Officer's Representative prior to placement to the lines and grades specified on the drawings and staked in the field. Any excess material shall be disposed of in accordance with Construction Specification 4, Channel Clearing and Shaping.
- 4) Except for the top 6 inches and around utilities, the maximum rock particles size shall be 6 inches and the maximum lift thickness shall be 12 inches. The top 6 inches shall be fine grain soil material suitable for establishing a seedbed. The maximum rock particle size in the top 6 inches and around utilities shall be 2 inches. The top 6 inches may be placed in one 6-inch lift. The maximum lift thickness around utilities shall be 4 inches.
- 5) The moisture content of the earth fill shall be such that a dense compact fill can be obtained. The moisture content shall be sufficient to permit molding a firm ball when firmly squeezed by hand. The earth fill shall not be so wet that water runs out of the ball when squeezed nor so dry that the ball easily crumbles when slightly deformed.
- 6) Compaction shall be Class C as defined in Section F above. Compaction shall be obtained by traversing the entire surface of each lift four (4) times with the track of a crawler type tractor weighing not less than 15 tons. Fill placed around any utilities shall be hand compacted. Scarify the previous lift layer prior to placement of more fill material.
- 7) The contractor shall exercise caution when placing and compacting fill around utilities. Utility lines will be bedded in 2 feet of fine grain fill. No rocks larger than 2 inches in diameter shall be in contact with a utility line.

b. Subsidiary Item, Earthfill

- 1) This item shall consist of providing any and all materials, equipment, and labor necessary to provide Earthfill to the lines and grades specified on the drawings.
- 2) In Section I, Measurement and Payment, shall not apply. No separate payment shall be made for Earthfill. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 24 — Drainfill

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing, placing, and compacting drainfill required in the construction of structure drainage systems.

B. Material

1. Method 1. Drainfill material must conform to the requirements of Material Specification 521, Aggregates for Drainfill and Filters. At a minimum of 30 days before delivery of materials to the site, the contractor must inform the engineer, in writing, of the source(s) from which drainfill material will be obtained. The contractor must provide the engineer free access to the source(s) for the purpose of obtaining samples for testing.
2. Method 2. Drainfill material must be sand, gravel, or crushed stone (or mixtures thereof) obtained from the specified sources. Select the material as necessary to avoid the inclusion of organic matter, clay balls, excessive fine particles, or other substances that would interfere with its free-draining properties.

C. Base Preparation

1. Foundation surface and trenches must be clean and free of organic matter, loose soil, foreign substances, and standing water when the drainfill is placed. Earth surfaces upon or against which drainfill will be placed must not be scarified.

D. Placement

1. Drainfill must not be placed until the subgrade has been inspected and approved by the engineer. Drainfill must not be placed over or around pipe or drain tile until the installation of the pipe or tile has been inspected and approved.
2. Place drainfill uniformly in layers not exceeding 12 inches in thickness before compaction. When compaction is accomplished by manually controlled equipment, the layers must not exceed 8 inches in thickness. Place the material to avoid segregation of particle sizes and to ensure the continuity and integrity of all zones. No foreign material must be allowed to become intermixed with or otherwise contaminate the drainfill.
3. Traffic must not be permitted to cross over drains at random. Equipment cross overs must be maintained and the number and location of such cross overs must be established and approved before beginning drainfill placement. Each cross over must be cleaned of all contaminating material and must be inspected and approved by the engineer before the placement of additional drainfill material.

4. Any damage to the foundation surface, trench sides, or bottom occurring during placement of drainfill must be repaired before drainfill placement is continued.
5. The upper surface of drainfill constructed concurrently with adjacent zones of earthfill must be maintained at a minimum elevation of 1 foot above the upper surface of adjacent earthfill.
6. Place drainfill over and around pipe or drain tile to avoid any displacement in line or grade of the pipe or tile.
7. Drainfill must not be placed adjacent to structures until the concrete has attained the strength specified in Section I of this specification. Determine the strength by compression testing of concrete test cylinders cast and field cured at the project site in accordance with ASTM C31 for determining when a structure may be placed into service.
8. When the required strength of the concrete is not specified as described above, placement of drainfill adjacent to concrete structures must not commence until the time intervals specified in figure 24-1 have elapsed following placement of the concrete.

Figure 24-1 Elapsed Time Before Fill Can be Place Near Concrete Structures

Structure Type	Time Interval (Days)
Vertical or near-vertical wall with earth loading on one side only (retaining walls and counterforts)	14
Walls backfilled on both sides simultaneously	7
Conduits and galleries, cast-in-place (with inside forms in place) (inside forms removed)	7 14
Conduits, precast, cradled	2
Conduits, precast, bedded	1
Cantilever outlet bents backfilled on both sides simultaneously	3

E. Control of Moisture

1. Control the moisture content of drainfill material as specified in Section I of this specification. When additional water is required, apply it in a manner to avoid excessive wetting to adjacent earthfill. Except as specified in Section I of this specification, control of moisture content will not be required.

F. Compaction

1. All Items. Compact drainfill according to the following requirements for the class of compaction specified. A pass in this section is defined as at least one complete coverage of the roller wheel, tire, or drum over the entire surface for each layer.
2. Class A Compaction. For drainfill materials with more than 70 percent passing the 3/4-inch sieve, compact each layer of drainfill to the minimum dry density specified in Section I of this specification as determined by ASTM D698. For drainfill materials with 70 percent or less passing the 3/4-inch sieve, compact each layer of drainfill to a relative density of not less than 70 percent as determined by ASTM D4254.

3. Class I Compaction. Compact each layer of drainfill by making a minimum of two passes over the entire surface with a steel drum vibrating roller weighing at least 5 tons and exerting a vertical vibrating force of not less than 20,000 pounds at a minimum frequency of 1,200 times per minute or by an approved equivalent method.
4. Class II Compaction. Compact each layer of drainfill by one of the following methods or by an approved equivalent method.
 - a. A minimum of two passes over the entire surface with a pneumatic-tired roller exerting a minimum pressure of 75 pounds per square inch.
 - b. A minimum of four passes over the entire surface with the track of a crawler-type tractor weighing at least 20 tons.
 - c. Controlled movement of the hauling equipment so that the entire surface is traversed by not less than one tread track of the loaded hauling equipment.
5. Class III Compaction. No compaction will be required beyond that resulting from the placing and spreading operations.
6. Class I and II. When compaction other than Class III compaction is specified, material placed in trenches or other locations inaccessible to heavy equipment must be compacted by manually controlled pneumatic or vibrating tampers as specified in Section I of this specification.
7. Heavy Equipment Limits. Heavy equipment must not be operated within 2 feet of any structure. Vibrating rollers must not be operated within 5 feet of any structure. Compaction by means of drop weights operating from cranes, hoists, or similar equipment will not be permitted.

G. Testing

1. The contractor must conduct such tests as necessary to verify that the drainfill material and the in-place drainfill meets the specification requirements.
2. The engineer must be granted access to perform such tests as are required to verify that the drainfill materials and the drainfill in place meets the requirements of the specifications. These tests are not intended to provide the contractor with information needed to ensure that the materials and workmanship meet the specification requirements. These verification tests will not relieve the contractor of the responsibility of performing required tests for that purpose.

H. Measurement and Payment

1. Method 1.
 - a. For items of work for which specific unit prices are established in the contract, the volume of drainfill within the neat lines shown on the drawings are measured and computed to the nearest cubic yard. Where the engineer directs placement of drainfill outside the neat lines to replace unsuitable foundation material, the volume of such drainfill is included. The volume included is only to the extent that the unsuitable condition is not a result of the contractor's improper construction operation as determined by the engineer.

- b. Payment for drainfill is made at the contract unit price for each type of drainfill, complete and in place. Except as otherwise specified in Section I of this specification, such payment will constitute full compensation for all labor, equipment, material, and other items necessary and incidental to the performance of the work.
- 2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of drainfill placed within the specified limits is computed to the nearest 0.1 ton by actual weight. Where the engineer directs placement of drainfill outside the neat lines to replace unsuitable foundation material, the weight of such drainfill is included. The weight included is only to the extent that the unsuitable condition is not a result of the contractor's improper construction operation as determined by the engineer.
- 3. Exceptions. Payment for drainfill is made at the contract unit price for each type of drainfill, complete and in place. Except as otherwise specified in Section I of this specification, such payment will constitute full compensation for all labor, equipment, material, and other items necessary and incidental to the performance of the work.
- 4. All Methods. Compensation for any item of work described in the contract but not included in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section I of this specification.

I. Items of Work and Construction Details

Items of work to be performed in conformance with this specification and construction details therefore are:

1. Subsidiary Item, Drainfill

- a. This item shall consist of furnishing and placing drainfill as shown on the drawings.
- b. Drainfill gradation shall meet ASTM C33, No. 57 coarse aggregate.
- c. Compaction shall be Class III.
- d. Section H, Measurement and Payment, shall not apply. No separate payment shall be made for Drainfill. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2, as applicable.**

Construction Specification 25 — Rockfill

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the construction of rockfill zones of embankments and other rockfills required by the drawings and specifications, including bedding where specified.

B. Material

1. Obtain material for rockfill and bedding from the specified sources unless otherwise specified in Section J of this specification. The material must be excavated, selected, processed, and handled as necessary to conform to the specified gradation requirements.

C. Foundation Preparation

1. Strip foundations for rockfill to remove vegetation and other unsuitable material or excavate as specified.
2. Except as otherwise specified, grade earth foundation surfaces to remove surface irregularities and test pits or fill other cavities with compacted earthfill of approximately the same kind and density as the adjacent foundation material.
3. Clean rock foundation surfaces of all loose material not conforming to the specifications for the rockfill.
4. Prepare abutments for rockfill zones of embankments as specified above for foundations.
5. Do not place rockfill or bedding until the foundation preparation is completed and the foundation and excavations have been inspected and approved.

D. Bedding

1. When a bedding layer beneath rockfill is specified, spread the bedding material uniformly on the prepared subgrade surfaces to the depth indicated. Compaction of the bedding material must be as specified in Section J of this specification.

E. Placement

1. Method 1. The rock must be dumped and spread into position in approximately horizontal layers not to exceed 3 feet in thickness. It must be placed to produce a reasonably homogeneous, stable fill that contains no segregated pockets of large

or small fragments or large, unfilled spaces caused by bridging of the larger rock fragments.

2. Method 2. The rock must be dumped and spread into position in approximately horizontal layers not to exceed 3 feet in thickness. The rock must be placed so that the completed fill is graded with the smaller rock fragments placed in the inner portion of the embankment and the larger rock fragments placed on the outer slopes. Place rock to produce a stable fill that contains no large, unfilled spaces caused by bridging of the larger fraction.

F. Control of Moisture

1. Control the moisture content of rockfill material as specified in Section J of this specification. When the addition of water is required, apply it in a manner that avoids excessive wetting of adjacent earthfill. Except as specified in Section J of this specification, control of the moisture content is not required.
2. Control the moisture content of the bedding material to ensure that bulking of the sand materials does not occur during compaction operations.

G. Compaction of Rockfill

1. Compact rockfill as described below for the class of compaction specified or by an approved equivalent method.
 - a. Class I Compaction. Compact each layer of fill by at least four passes over the entire surface with a steel drum vibrating roller that weighs at least 5 tons and exerts a vertical vibrating force of not less than 20,000 pounds at a frequency not less than 1,200 times per minute.
 - b. Class II Compaction. Compact each layer of fill by at least four passes over the entire surface by a track of a crawler-type tractor weighing at least 20 tons.
 - c. Class III Compaction. No compaction is required beyond that resulting from the placing and spreading operations.
2. Heavy equipment must not be operated within 2 feet of any structure. Vibrating rollers must not be operated within 5 feet of any structure. Compaction by means of drop weights operating from a crane, hoist, or similar equipment is not permitted.
3. When compaction other than Class III compaction is specified, rockfill placed in trenches or other locations inaccessible to heavy equipment must be compacted by manually controlled pneumatic or vibrating tampers or by equivalent methods approved by the engineer.

H. Compaction of Bedding

1. Compact bedding according to the following requirements for the class of compaction specified:
 - a. Class A Compaction. Compact each layer of bedding to a relative density of not less than 70 percent as determined by ASTM D4254.
 - b. Class I Compaction. Compact each layer of bedding by at least two passes over the entire surface with a steel drum vibrating roller weighing at least 5

tons and exerting a vertical vibrating force not less than 20,000 pounds at a frequency not less than 1,200 times per minute or by an approved equivalent method. A pass is defined as at least one passage of the roller wheel, track, tire, or drum over the entire surface of the bedding layer

- c. Class II Compaction. Compact each layer of bedding by one of the following methods or an equivalent method approved by the engineer.
 - (1) At least two passes over the entire surface with a pneumatic, rubber-tired roller exerting a minimum pressure of 75 pounds per square inch. A pass is defined as at least one passage of the roller wheel, track, tire, or drum over the entire surface of the bedding layer.
 - (2) At least four passes over the entire surface with the track of a crawler-type tractor weighing a minimum of 20 tons.
 - (3) Controlled movement of the hauling equipment so that the entire surface is traversed by a minimum of one tread track of the loaded equipment.
 - d. Class III Compaction. No compaction is required beyond that resulting from the placing and spreading operations.
- 2. Heavy equipment must not be operated within 2 feet of any structure. Vibrating rollers must not be operated within 5 feet of any structure. Compaction by means of drop weights operating from a crane, hoist, or similar equipment is not permitted.
 - 3. When compaction other than Class III is specified, bedding placed in trenches or other locations inaccessible to heavy equipment must be compacted by manually controlled pneumatic or vibrating tampers or by equivalent methods approved by the engineer.

I. Measurement and Payment

- 1. For items of work for which specific unit prices are established in the contract, the volume of each type of rockfill, including bedding, with the zone boundaries and limits specified on the drawings or established by the engineer, is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- 2. Payment for each type of rockfill is made at the contract unit price for that type of fill. Except as otherwise specified in Section J of this specification, such payment will constitute full compensation for all labor, equipment, material, and other items necessary and incidental to the performance of the work, including furnishing, placing, and compacting the bedding material.
- 3. Compensation for any type of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section J of this specification.

J. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details are:

a. Rockfill, General Requirements

- 1) Excavated deposition material that will be used for Rockfill shall be approved by the Contracting Officer's Representative prior to placement to the lines and grades specified on the drawings and staked in the field. Any excess material shall be disposed of in accordance with Construction Specification 4, Channel Clearing and Shaping.
- 2) Rockfill shall consist of clean durable rock. Rockfill shall contain no more than 10 percent of material, by visual examination, smaller than 3 inches. The maximum particle size shall be 12 inches, as measured in the maximum dimension. Rockfill shall be classified as moderately hard to hard according to the ASCE Rock Classification System.
- 3) In Section E, Placement, Method 1 shall apply except that the layer thickness shall not be more than 12 inches.
- 4) Moisture Control is not required for Rockfill.
- 5) Rockfill shall be compacted by use of onsite heavy equipment with a minimum of 3 passes. Additional passes may be required to produce a condition of non-movement under compaction equipment.
- 6) Rockfill shall not be placed and compacted around utilities unless otherwise directed by the Contracting Officer's Representative.

b. Subsidiary Item, Rockfill

- 1) This item shall consist of providing any and all materials, equipment, and labor necessary to provide Rockfill to the lines and grades specified on the drawings.
- 2) In Section I, Measurement and Payment, shall not apply. No separate payment shall be made for Rockfill. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 34 — Steel Reinforcement

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing and placing steel reinforcement for reinforced concrete or pneumatically applied mortar.

B. Material

1. Steel reinforcement must conform to the requirements of Material Specification 539, Steel Reinforcement (for concrete). Before reinforcement is placed, the surface of the bars and fabric and any metal supports must be cleaned to remove any loose, flaky rust, mill scale, oil, grease, or other undesirable coatings or foreign substances. Epoxy-coated steel reinforcement must be free of surface damage. After placement, the reinforcement must be maintained in a clean and serviceable condition until it is completely embedded within the concrete.

C. Bar schedule, lists and diagrams

1. Any supplemental bar schedules, bar lists, or bar-bending diagrams required in section J of this specification for the fabrication and placement of steel reinforcement must be provided by the contractor. Before reinforcement is placed, the contractor must furnish four copies of any such lists or diagrams to the contracting officer for approval. Acceptance of the reinforcement is not based on approval of these lists or diagrams but on inspection of the steel reinforcement after it has been placed, tied, and supported and is ready to receive concrete.

D. Bending

1. Reinforcement must be cut and bent in compliance with the requirements of the American Concrete Institute (ACI) Standard 315. Bars must not be bent or straightened in a manner that will injure or weaken the material. Bars with kinks, cracks, or improper bends must be rejected.

E. Splicing bar reinforcement

1. Method 1. Splices of reinforcement may only be made at locations shown on the drawings and provided by the steel schedule. Placement of bars at the lap splice locations shown, when not in contact, must not be farther apart than one-fifth the shown lap length and, in all cases, no greater than 6 inches.
2. Method 2. Splices of reinforcement must be limited to those locations shown on the drawings. Splice lengths must be determined before fabrication and meet the requirements of ACI Standard 318, Building Code Requirements for Reinforced

Concrete, based on design information in section J of this specification. Bar placement drawings and schedules must be provided for approval before fabrication. The drawings must show all splice locations, layouts, and lap dimensions.

F. Splicing welded wire reinforcement

1. Unless otherwise specified, welded wire reinforcement must be spliced in the following manner:
 - a. End-to-end. Adjacent sections must be spliced end-to-end (longitudinal lap) by overlapping a minimum of one full mesh plus 2 inches plus the length of the two end overhangs. The splice length is measured from the end of the longitudinal wires in one piece of fabric to the end of the longitudinal wire in the lapped piece of fabric.
 - b. Side-to-side. Adjacent sections must be spliced side-to-side (transverse lap) a minimum of one full mesh plus 2 inches. The splice length must be measured from the centerline of the first longitudinal wire in one piece of fabric to the centerline of the first longitudinal wire in the lapped piece of fabric.

G. Placing

1. Reinforcement must be accurately placed and secured in position to prevent its displacement during the placement of concrete. Tack welding of bars is not permitted. Metal chairs, metal hangers, metal spacers, and concrete chairs may be used to support the reinforcement. Metal hangers, spacers, and ties must be placed in such a manner that they are not exposed in the finished concrete surface. The legs of metal chairs or side form spacers that may be exposed on any face of slabs, walls, beams, or other concrete surfaces must have a protective coating or finish. The coating or finish must be hot dip galvanized, epoxy coated, plastic coated, or stainless steel. Metal chairs and spacers not fully covered by a protective coating or finish must have a minimum cover of 0.75 inch of concrete over the unprotected metal part. The exception is that those with plastic coatings may have a minimum cover of 0.5 inch of concrete over the unprotected metal part. Precast concrete chairs must be manufactured of the same class of concrete as specified for the structure and must have the tie wires securely anchored in the chair or a V-shaped groove at least 0.75 inch in depth molded into the upper surface to receive the steel bar at the point of support. Precast concrete chairs must be clean and moist at the time concrete is placed.
2. High-density or structural plastic rebar accessories designed to ensure maximum concrete bond may be substituted for metal or concrete accessories in spacer applications as approved by the contracting officer. Exposure of plastic rebar accessories at the finished concrete surface must be kept to a minimum. Plastic rebar accessories, when used, must be staggered along adjacent parallel bars and placed at intervals no closer than 12 inches. Plastic rebar accessories must not be used in concrete sections 6 inches or less in thickness.

3. Reinforcement must not be placed until the prepared site has been inspected and approved. After placement of the reinforcement, concrete must not be placed until the reinforcement has been inspected and approved by the responsible engineer.

H. Storage

1. Steel reinforcement stored at the work site must be placed on platforms, skids, or other supports. This is done to avoid contact with the ground and to protect the material from mechanical damage and corrosion.

I. Measurement and payment

1. Method 1.

- a. For items of work with specific unit prices established in the contract, the weight of steel reinforcement placed in the concrete in accordance with the drawings is determined to the nearest pound by computation from the placing drawings. Measurement of hooks and bends is based on the requirements of ACI Standard 315. Computation of weights of reinforcement is based on the unit weights established in figures 34–1 and 34–2 of this specification. Computation of weights for welded wire reinforcement not shown in figure 34–2 must be based on ACI Standard 315. The area of welded wire reinforcement placed in the concrete in accordance with the drawings is determined to the nearest square foot by computation from the placing drawings with no allowance for required laps. The weight of steel reinforcing in extra splices or extra-length splices approved for the convenience of the contractor or the weight of supports and ties is not included in the measurement for payment.
- b. Payment for furnishing and placing reinforcing steel is made at the contract unit price. Such payment constitutes full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work including preparing and furnishing bar schedules, lists, or diagrams; furnishing and attaching ties and supports; and furnishing, transporting, storing, cutting, bending, cleaning, and securing all reinforcements.

2. Method 2.

- a. For items of work with specific unit prices established in the contract, the weight of bar reinforcement placed in the concrete in accordance with the drawings is determined to the nearest pound by computation from the placing drawings. Measurement of hooks and bends is based on the requirements of ACI Standard 315. Computation of weights of bar reinforcement is based on the unit weights established in figure 34–1 of this specification. The weight of steel reinforcing in extra splices or extra-length splices approved for the convenience of the contractor or the weight of supports and ties is not included in the measurement for payment.
- b. The area of welded wire reinforcement placed in the concrete in accordance with the drawings is determined to the nearest square foot by computation from the placing drawings with no allowance for required laps.

- c. Payment for furnishing and placing bar reinforcing steel is made at the contract unit price for bar reinforcement. Payment for furnishing and placing welded wire reinforcing steel is made at the contract unit price for welded wire reinforcement. Such payment constitutes full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work including preparing and furnishing bar schedules, lists, or diagrams; furnishing and attaching ties and supports; and furnishing, transporting, cutting, bending, cleaning, and securing all reinforcement.
3. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items to which they are made subsidiary are identified in section J of this specification.

Figure 34–1 Standard reinforcing bars

Bar Size (English)¹	Bar Size (Metric)²	Weight (lb/ft)
3	10	0.376
4	13	0.668
5	16	1.043
6	19	1.502
7	22	2.044
8	25	2.670
9	29	3.400
10	32	4.303
11	36	5.313
14	43	7.650
18	57	13.600

1/ The bar diameter (inches) equals the bar size number divided by eight. For example, the diameter of a #4 bar is $4 \div 8 = 0.5$ inch.

2/ The metric bar size has been rounded to a whole number that represents the approximate diameter of the bar in millimeters.

Figure 34–2 Rectangular welded wire reinforcement

Style designation - W-number^{1,2}	Style designation - steel wire gauge (former designation)^{1,2}	Weight (lb/100 ft²)
6 × 6 – W1.4 × W1.4	6 × 6 – 10 × 10	21
6 × 6 – W2.1 × W2.1	6 × 6 – 8 × 8	30
6 × 6 – W2.9 × W2.9	6 × 6 – 6 × 6	42
6 × 6 – W4.0 × W4.0	6 × 6 – 4 × 4	58
4 × 4 – W1.4 × W1.4	4 × 4 – 10 × 10	31
4 × 4 – W2.1 × W2.1	4 × 4 – 8 × 8	44
4 × 4 – W2.9 × W2.9	4 × 4 – 6 × 6	62
4 × 4 – W4.0 × W4.0	4 × 4 – 4 × 4	85
4 × 12 – W2.1 × W0.9 ^{2/}	4 × 12 – 8 × 12	25
4 × 12 – W2.5 × W1.1 ^{2/}	4 × 12 – 7 × 11	31

1/ Style designation is defined in ACI Standard 315 of the American Concrete Institute.

2/ Welded smooth wire reinforcement with wires smaller than size W1.4 is manufactured from galvanized wire.

J. Items of work and construction details

Items of work to be performed according to this specification and the construction details are:

1. General

- a. This item shall consist of providing the reinforcement bars, drilling the bedrock, and installing the reinforcement bars as shown on the drawings and staked in the field.
- b. The NRCS representative in the field shall determine the extent of the rebar anchoring needed.

2. Subsidiary Item, Steel Reinforcement, Rebar Anchoring for Grouted Rock Riprap on Bedrock

- a. Steel reinforcement shall be #8 bars, Grade 60 minimum.
- b. The drill hole diameter shall be 1-1/4 inches to the depth specified on the drawings.
- c. In Section I, Measurement and Payment, shall not apply. No separate payment shall be made for Steel Reinforcement. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1.**

3. Bid Item 3B, Steel Reinforcement, Foundation Preparation on Shallow Bedrock (Rebar Anchoring)

- a. Steel reinforcement shall be #10 bars, Grade 60 minimum.
- b. The drill hole diameter shall be 1-3/8 inches to the depth and spacing specified on the drawings.
- c. Epoxy or grouting of the reinforcement bars is not required.
- d. In Section I, Measurement and Payment, shall not apply. Payment for Bid Item 3, Steel Reinforcement, Foundation Preparation on Shallow Bedrock (Rebar Anchoring) will be made at the contract unit price of square yard of foundation preparation. The payment extents shall be the footprint of the gabion wall base that requires rebar placement. The FAR 52.211-18 Variation in Estimated Quantities clause does not apply. The estimated quantity is only an estimate and does not guarantee the quantity and this method will be used at the site.

Construction Specification 45 — Plastic Pipe

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing and installing plastic pipe (except corrugated polyethylene pipe) and the necessary fittings and appurtenances as shown on the drawings or as specified herein.

B. Material

1. Pipe, fittings, and gaskets must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
2. Perforated pipe must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
3. Unless otherwise specified, concrete must conform to the requirements of Construction Specification 32, Structure Concrete, and Section H of this specification.
4. Unless otherwise specified, earth backfill must conform to the requirements of Construction Specification 23, Earthfill.
5. Unless otherwise specified, drainfill must conform to the requirements of Construction Specification 24, Drainfill.

C. Handling and Storage

1. Deliver pipe to the job site and handled by means that provide adequate support to the pipe and do not subject it to undue stresses or damage. When handling and placing plastic pipe, take care to prevent impact blows, abrasion damage, and gouging or cutting by metal edges, surfaces or rocks. The manufacturer's special handling requirements must be strictly observed. Special care must be taken to avoid impact when the pipe must be handled at a temperature of 40 degrees Fahrenheit or less.
2. Store pipe on a relatively flat surface so that the barrels are evenly supported. Unless the pipe is specifically manufactured to withstand exposure to ultraviolet radiation, cover it with an opaque material when stored outdoors for 15 days or longer.

D. Excavation

1. Excavate in accordance with Construction Specification 21, Excavation, and Section N of this specification or as shown on the drawings.
2. Excavate the pipe foundation a minimum of 4 inches lower than the pipe grade shown on the drawings or staked in the field whenever bedrock, boulders, cobbles, or other material that may cause pipe damage is encountered at planned pipe grade.

E. Laying the Pipe

1. Install plastic pipe conduits complete with fittings and other related appurtenances to the lines and grades shown on the drawings or specified in Section N of this specification. Install the pipe so that there is no reversal of grade between joints unless otherwise shown on the drawings. The pipe must not be dropped or dumped on the bedding or into the pipe trench. The ground surface near the pipe trench must be free of loose rocks and stones greater than 1 inch in diameter. This ensures that rock will not be displaced and impact the pipe.
2. Just before placement, inspect each pipe section to ensure that all foreign material is removed from inside the pipe. The pipe ends and the couplings must be free of foreign material when assembled. At the completion of a work shift, all open ends of the pipeline must be temporarily closed off using a suitable cover or plug.
3. Take care to prevent distortion and damage during hot or cold weather. During unusually hot weather (daytime high temperature of more than 90 °F), lightly backfill the pipe assembled in the trench or shade to keep it as near to ground temperature as possible until final backfill is placed. Backfill operations must be performed during daily construction periods when the ground temperature and the temperature of the pipe do not vary more than 40 degrees Fahrenheit.
4. Install perforated pipe with the perforations down and oriented symmetrically about the vertical centerline. Perforations must be clear of any obstructions on the inside and outside of the pipe when the pipe is approved by the engineer for backfill.
5. During installation, firmly and uniformly bed the pipe throughout its entire length to the depth and in the manner specified in Section N of this specification or as shown on the drawings. Place bell-holes in bedding material under bells, couplings, and other fittings to ensure the pipe is uniformly supported throughout its entire length. Blocking or mounding beneath the pipe to bring the pipe to final grade is not permitted.

F. Pipe Embedment

1. Earth bedding.
 - a. Firmly and uniformly place the pipe on compacted earthfill bedding or an in-place earth material bedding of ample bearing strength to support the pipe without noticeable settlement. The earth material on which the pipe is placed must have uniform density to prevent differential settlement.

- b. Unless otherwise specified, form a groove that closely conforms to the outside surface of the pipe in the bedding. The depth of the groove must be equal to or greater than 0.3 of the pipe diameter.
 - c. Compact earth bedding to a density not less than adjacent, undisturbed, in-place earth material or be compacted earth backfill. Earthfill material used for compacted earth bedding must be free of rocks or stones greater than 1 inch in diameter and earth clods greater than 2 inches in diameter. The pipe must be loaded sufficiently during the compaction of bedding under the haunches and around the sides of the pipe to prevent displacement from its final approved placement.
2. Sand, gravel, or crushed rock bedding. When sand, gravel, or crushed rock bedding is specified, firmly and uniformly place the pipe on the bedding material. Material for bedding must not exceed 1 inch in diameter. Unless otherwise specified in Section N of this specification or shown on the drawings, the coarse-grained bedding material must be carefully placed and compacted to a depth equal to or greater than 0.3 of the diameter of the pipe above the bottom of the pipe. The pipe must be loaded sufficiently during backfilling and compaction around the sides to prevent displacement of the pipe from its final approved placement.
 3. Pipe encased in drainfill. Firmly and uniformly place the pipe on bedding of specified drainfill, which must be placed and compacted as specified in Section N of this specification or as shown on the drawings to form a continuous, uniform support around the entire circumference of the pipe. The pipe must be loaded sufficiently during backfilling around the sides and during compaction to prevent displacement of the pipe.

G. Backfill

1. Initial backfill.
 - a. Unless otherwise specified, initial backfill to 6 inches above the top of the conduit is required. Earth haunching and initial backfill material must consist of soil material free of rocks, stones, or hard clods more than 1 inch in diameter. Coarse backfill material must be the specified sand, gravel, crushed rock, or drainfill material.
 - b. Initial backfill must be placed in two stages. In the first stage (haunching), backfill is placed to the pipe spring line (center of pipe). In the second stage, it is placed to 6 inches above the top of the pipe.
 - c. The first stage material must be worked carefully under the haunches of the pipe to provide continuous support throughout the entire pipe length. The haunching backfill material must be placed in layers that have a maximum thickness of about 6 inches and are compacted as specified in Section N of this specification or as shown on the drawings. During compaction operations, care must be taken to ensure that the tamping or vibratory equipment does not come in contact with the pipe, and the pipe is not deformed or displaced.
 - d. When pressure testing is not specified, cover the pipe with a minimum of 6 inches of backfill material as soon as possible following assembly of the pipe in the trench but not later than within the same day that placement has

occurred. When pressure testing is specified, place sufficient backfill material over the pipe to anchor the conduit against movement during pressure testing activities.

2. Final backfill.

- a. Final backfill consists of placing the remaining material required to complete the backfill from the top of the initial backfill to the ground surface, including mounding at the top of the trench. Final backfill material within 2 feet of the top of the pipe must be free of debris or rocks larger than 3 inches nominal diameter. Coarse backfill material must be the specified sand, gravel, crushed rock, or drainfill. Final backfill must be placed in approximately uniform, compacted layers. Final backfill compaction requirements must be as specified in Section N of this specification or as shown on the drawings.
- b. Vehicles or construction equipment must not be allowed to cross the pipe until the minimum earth cover and required density as specified in Section 14 of this specification have been obtained.

H. Pipe Encasement in Concrete

1. Concrete encasement must be carefully placed to form a continuous, uniform support around the entire circumference of the pipe as specified in Section N of this specification or as shown on the drawings. Pipes encased in concrete must be securely anchored to prevent movement of the pipe during concrete placement. A clear distance of 1.5 inch must be maintained between the pipe and the reinforcing steel.
2. The concrete for the encasement must conform to the requirements of Construction Specification 32, Structure Concrete, for Class 3000M concrete unless otherwise specified.

I. Joints

1. Unless otherwise specified in Section N of this specification or shown on the drawings, joints must be either bell-and-spigot types with elastomeric gaskets, coupling types, solvent cement bell and spigot, or jointed by butt heat fusion. When a lubricant is required to facilitate joint assembly, it must be a type having no deleterious effect on the gasket or pipe material.
2. Pipe joints must be watertight at the pressures specified except where unsealed joints are indicated.
3. Install pipe and join in accordance with the manufacturer's recommendations. Laying deflections and joint fitting or stab depths must be within the manufacturer's recommended tolerances.
4. When solvent cement joints are specified for PVC or ABS pipe and fittings, they must be made in accordance with the following ASTMs and the related appendix of each ASTM: D2855 for PVC pipe and fittings and D2235 for ABS pipe and fittings.

5. Flanged, banded, heat-fusion, or elastomeric-sealed mechanical joints must be used when joining polyethylene (PE) and high-density polyethylene (HDPE) pipe and fittings unless otherwise specified in Section N of this specification or as shown on the drawings.
6. Cut pipe ends square and debur to provide a uniform, smooth surface for the jointing process. Place reference marks on the spigot ends to assist in determining when proper seating depth has been achieved within the joint.

J. Fittings

1. Unless otherwise specified, steel fittings, valves, and bolted connections must be painted or coated as recommended by the manufacturer.
2. Fittings for non-pressure pipe must be of the same or similar material as the pipe and must provide the same durability, watertightness, and strength as the pipe unless otherwise specified.

K. Thrust Blocks and Anchors

1. When specified, install concrete thrust blocks and anchors as shown on the drawings or as specified in Section N of this specification.
2. The concrete for thrust blocks and anchors must conform to the requirements of Construction Specification 32, Structure Concrete, for Class 3000M concrete unless otherwise specified in Section N of this specification.
3. The thrust block cavity must be hand-dug into undisturbed soil or previously placed, compacted backfill. The cavity must be formed with soil or wood to hold the freshly placed concrete without displacement until an initial set has occurred.
4. When excavation beyond the designated trench widths and depths as shown on the drawings or specified in Section N of this specification occurs at locations where installation of concrete thrust blocks is required, the contractor must install an alternative thrust block provision. The concrete thrust block must have a thickness of one pipe diameter and a contact face area that must be formed against the pipe as shown on the drawings or as specified in Section N of this specification. Place backfill on all sides of the thrust block and to the sides of the excavation. It must be compacted as specified for initial backfill.

L. Pressure Testing

1. Method 1. Pressure testing of the completed conduit is not required.
2. Method 2. Test the conduit for leaks in the following manner:
 - a. Before pressure testing:
 - (1) Joints of the assembled pipeline must be allowed to cure as recommended by the manufacturer.
 - (2) Flush and clean the pipeline.
 - (3) Place all concrete anchors and thrust blocks and allowed to cure for a minimum of 3 days.

- (4) Earth backfill must be sufficient to anchor the conduit against movement during the pressure testing and must be compacted as specified in Section N of this specification or as shown on the drawings.
 - (5) The conduit must be braced, anchored, or both at each end to restrict all potential pipe movement.
 - (6) Plug the ends of the conduit. The upstream plug must have a standpipe installed vertically with a minimum diameter of 2 inches and must be equipped with a shutoff valve. All high points in the line must be vented to evacuate air pockets. The conduit and the standpipe must be slowly filled with water so no air is entrapped during the filling operation. After filling is complete, all vents must be closed.
 - b. During pressure testing, the water level in the standpipe must be continuously maintained at a minimum of 10 feet above the highest invert elevation of the conduit for no less than 2 hours unless otherwise specified in Section N of this specification or as shown on the drawings.
 - (1) The volume of water leakage in the 2-hour test period must be recorded. The maximum allowable leakage (L) in gallons per hour must not exceed 0.02 times the nominal pipe diameter (D) in inches for each 1,000 feet of pipeline, which is about 50 pipe joints ($L = 0.02 \times D$).
 - c. When observed leakage exceeds the allowable amount, leaks must be sealed by replacement of pipe and fittings as necessary. The conduit must be retested as described above. This procedure must be repeated until the conduit leakage does not exceed the allowable volume specified above.
 - (1) The contractor must be fully responsible for any and all work required to correct leakage exceeding the amount specified.
3. Method 3. Test the conduit for leaks in the following manner:
- a. Before pressure testing:
 - (1) Joints of the assembled pipeline must be allowed to cure as recommended by the manufacturer.
 - (2) Flush and clean the pipeline.
 - (3) All concrete anchor and thrust blocks must be in place and allowed to cure for at least 3 days.
 - (4) Earth backfill must be sufficient to anchor the conduit against movement during the pressure testing and compacted as specified in Section N of this specification or as shown on the drawings.
 - (5) The conduit must be braced or anchored at each end to prevent all potential pipe movement.
 - (6) Plug the ends of the conduit and attach a pressure gauge to the upstream and downstream ends. All high points along the pipeline must be vented to permit the complete removal of all air within the pipeline. Slowly fill the conduit with water so no air is entrapped during the filling operations.
 - b. The testing pressure specified in Section N of this specification must be continuously maintained at the upstream gauge for a minimum of 2 hours. The pressure at the downstream gauge must not exceed the pressure rating of the pipe.

- c. The volume of water leakage for the 2-hour test period must be recorded. Maximum allowable leakage must be in accordance with figure 45-1 or equation 45.1:

Figure 45-1 Allowable leakage for plastic pipe (gal/hr/1,000 feet, or 50 pipe joints)^{1/}

Nominal Pipe Size (in)	Test Pressure in the Pipeline (lb/in ²)			
	50	100	150	200
	-----Allowable Leakage -----			
4	.19	.27	.33	.38
6	.29	.41	.50	.57
8	.38	.54	.66	.76
10	.48	.68	.83	.96
12	.57	.81	.99	1.15
14	.67	.95	1.16	1.34
15	.72	1.02	1.25	1.44
16	.76	1.07	1.32	1.52
18	.86	1.22	1.49	1.72

1/ Computation basis

$$L = \frac{ND \sqrt{P}}{7,400}$$

Eq: 45.1

where:

L = allowable leakage in gallons per hour

N = number of joints (pipe and fittings)

D = nominal diameter of pipe in inches

P = test pressure in pounds per square inch

- d. When observed leakage exceeds the allowable volume, leaks must be sealed by replacement of pipe and fittings as necessary. The conduit must be retested as described in this section. The procedure must be repeated until the conduit leakage does not exceed the allowable amount specified above.
- e. The contractor must be fully responsible for any and all work required to correct leakage exceeding the amount specified.

M. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.

2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for the kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
3. Method 3. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract unit price for the kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
4. Method 4. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the pipe sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
5. Methods 3 and 4. Excavation, backfill, and bedding are paid separately under their respective bid items.
6. All Measurement and Payment Methods. Compensation for any items of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in Section N of this specification.

N. Items of Work and Construction Details

Items of work to be performed in conformance with this specification and construction details therefore are:

1. Subsidiary Item, Plastic Pipe, PVC Pipe-3 inch Diameter

- a. This item shall consist of furnishing and installing the 3-inch diameter PVC and appurtenances in the grouted rock riprap as shown on the drawings.
- b. Pipe shall consist of solid and perforated pipes. The locations and diameter of the perforations shall be as shown on the drawings.
- c. All pipe shall meet ASTM D1785, PVC Schedule 40. Pipe Fittings shall meet ASTM D2466.
- d. Section F, Pipe Embedment, shall not apply. The pipe shall be firmly encased in the rock riprap and drainfill as shown on the drawings. The pipe inlet shall be in contact with the geotextile prior to the placement of the concrete grout.
- e. Method 1 of Section L, Pressure Testing, shall apply.
- f. In Section M, Measurement and Payment, shall not apply. No separate payment shall be made for Plastic Pipe. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1.**

Construction Specification 61 — Rock Riprap

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of the construction of rock riprap revetments and blankets, including filter or bedding where specified.

B. Material

1. Rock riprap must conform to the requirements of Material Specification 523, Rock for Riprap, or, if so specified, must be obtained from designated sources. It must be free from dirt, clay, sand, rock fines, and other material not meeting the required gradation limits.
2. At least 30 days before rock is delivered from other than designated sources, the contractor must designate in writing the source from which rock material will be obtained and provide information satisfactory to the contracting officer that the material meets contract requirements. The contractor must provide the responsible engineer free access to the source for the purpose of obtaining samples for testing. The size and grading of the rock must be as specified in section H.
3. Rock from approved sources must be excavated, selected, and processed to meet the specified quality and grading requirements at the time the rock is installed.
4. Based on a specific gravity of 2.65 (typical of limestone and dolomite) and assuming the individual rock is shaped midway between a sphere and a cube, typical size/weight relationships are given in figure 61-1:

Figure 61-1 Typical riprap size/weight relationship

Sieve Size of Rock	Approx. Weight of Rock	Weight of Test Pile
16 inches	300 pounds	6,000 pounds
11 inches	100 pounds	2,000 pounds
6 inches	15 pounds	300 pounds

5. When specified in section H or when it is necessary to verify the gradation of the rock riprap, a particle size analysis must be performed in accordance with ASTM D5519, Test Method A or B. The analysis must be performed at the worksite on a test pile of representative rock. The mass of the test pile must be at least 20 times the mass of the largest rock in the pile. The results of the test are compared to the gradation required for the project. Test pile results that do not meet the construction specifications must be cause for the rock to be rejected. The test pile that meets contract requirements must be left on the jobsite as a sample for visual comparison. The test pile must be used as part of the last rock riprap to be placed.

6. Filter or bedding aggregates, when required, must conform to Material Specification 521, Aggregates for Drainfill and Filters, unless otherwise specified. Geotextiles must conform to Material Specification 592, Geotextile.

C. Subgrade preparation

1. The subgrade surface on which the rock riprap, filter, bedding, or geotextile is to be placed must be cut or filled and graded to the lines and grades shown on the drawings. When fill to subgrade lines is required, it must consist of approved material and conform to the requirements of the specified class of earthfill.
2. Rock riprap, filter, bedding, or geotextile must not be placed until the foundation preparation is completed and the subgrade surface has been inspected and approved.

D. Equipment-placed rock riprap

1. The rock riprap must be placed by equipment on the surface and to the depth specified. It must be installed to the full course thickness in one operation and in such a manner as to avoid serious displacement of the underlying material. The rock for riprap must be delivered and placed in a manner that ensures the riprap in place is reasonably homogeneous with the larger rocks uniformly distributed and firmly in contact one to another with the smaller rocks and spalls filling the voids between the larger rocks. Some hand placing may be required to provide a neat and uniform surface.
2. Rock riprap must be placed in a manner to prevent damage to structures. Hand placing is required as necessary to prevent damage to any new and existing structures.

E. Hand-placed rock riprap

1. The rock riprap must be placed by hand on the surface and to the depth specified. It must be securely bedded with the larger rocks firmly in contact one to another without bridging. Spaces between the larger rocks must be filled with smaller rocks and spalls. Smaller rocks must not be grouped as a substitute for larger rock. Flat slab rock must be laid on its vertical edge except where it is laid like paving stone and the thickness of the rock equals the specified depth of the riprap course.

F. Filter or bedding

1. When the contract specifies filter, bedding, or geotextile beneath the rock riprap, the designated material must be placed on the prepared subgrade surface as specified. Compaction of filter or bedding aggregate is not required, but the surface of such material must be finished reasonably smooth and free of mounds, dips, or windrows.

G. Measurement and payment

1. Method 1.
 - a. For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is

computed to the nearest ton by actual weight. The volume of each type of filter or bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas. For each load of rock riprap placed as specified, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

- b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

2. Method 2.

- a. For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest 0.1 ton by actual weight. The quantity of each type of filter or bedding aggregate delivered and placed within the specified limits is computed to the nearest 0.1 ton. For each load of rock riprap placed as specified, the contractor must furnish to the engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton. For each load of filter or bedding aggregate, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.
- b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

3. Method 3.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap and filter or bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

4. Method 4.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap, including filter and bedding aggregate, is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap, including filter and bedding. Such payment is considered full compensation for completion of the work.

5. Method 5.

- a. For items of work for which specific unit prices are established by the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest ton by actual weight. For each load of rock for riprap placed as specified, the contractor must furnish to the responsible

engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

- b. Payment is made at the contract unit price for each type of rock riprap, and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

6. Method 6.

- a. For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.
- b. Payment is made at the contract unit price for each type of rock riprap and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

7. All methods.

- a. The following provision applies to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section H.
- b. No separate payment is made for testing the gradation of the test pile. Compensation for testing is included in the appropriate bid item for riprap.

H. Items of work and construction details

Items of work to be performed in conformance with this specification and details are:

1. Bid Item 4B, Rock Riprap, Riprap Foundation Protection, Site 2

- a. This item shall consist of furnishing and installing rock riprap for the gabion basket foundation preparation or other areas as shown on the drawings or staked in the field. Use of this item will be at the discretion of the NRCS Engineer or Inspector in the field.
- b. In Section B, Material, rock shall meet the requirements of Rock Type I, Material Specification 523, or be obtained from sources with current approval from the WV DOT. The contractor shall provide a gradation test as specified with the material certification prior to the delivery of rock riprap to the site.
- c. The statement in Section B.2 that begins with “At least 30 days before rock is delivered...” is to be changed to read “At least 7 days before rock is delivered...”.
- d. Rock riprap shall be equipment placed.
- e. The rock riprap shall be well-graded between the nominal size of 6 and 18 inches. Well graded shall mean that a broad range of rock particle sizes are represented and that substantial amounts of each rock particle sizes are included. Gabion stone is to be “chinked” on the surface of the rock riprap foundation so that there is a smooth surface on which to construct the gabions. The gabion stone size shall conform to the requirements in Section B of Construction Specification 64, Gabions and Gabion Mattresses.
- f. Method 5 of Section G, Measurement and Payment, shall apply. The FAR 52.211-18 Variation in Estimated Quantity clause does not apply. The estimated quantity is only an estimate and does not guarantee the quantity and this method will be used at the site.

2. Subsidiary Item, Rock Riprap

- a. This item shall consist of providing any and all materials, equipment, and labor necessary to provide Rock Riprap as shown on the drawings in locations other than Gabion basket foundation.
- b. In section G, Measurement and Payment, shall not apply. No separate payment will be made for Rock Riprap. Compensation for this item is subsidiary to and included in the payment for **Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 62 — Gouted Rock Riprap

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing, transporting, and installing grouted rock riprap revetments and blankets, including filters or bedding where specified.

B. Material

1. Rock for riprap must conform to the requirements of Material Specification 523 or, if so specified, must be obtained from designated sources. It must be free from dirt, clay, sand, rock fines, and other material not meeting the required gradation limits.
 - a. At least 30 days before rock is delivered from other-than-designated sources, the contractor must designate, in writing, the source from which rock material will be obtained and provide information satisfactory to the engineer that the material meets contract requirements. The contractor must provide the engineer free access to the source for the purpose of obtaining samples for testing. The size and grading of the rock must be as specified in Section M of this specification.
 - b. Rock from approved sources must be excavated, selected, and processed to meet the specified quality and grading requirements at the time the rock is installed.
 - c. When specified in Section M of this specification or when requested by the contracting officer, a gradation quality-control check must be made by the contractor and subject to inspection by the engineer. The test must be performed at the work site in accordance to ASTM D5519 Test Method B, Size, Size-Range Grading, on a test pile of representative rock. The weight or size of the test pile must be large enough to ensure a representative gradation of rock from the source and provide test results within a 5 percent accuracy.
 - d. Based on a specific gravity of 2.65 (typical of limestone and dolomite) and assuming the individual rock is shaped midway between a sphere and a cube, typical size/weight relationships are given in figure 62-1:

Figure 62-1 Typical riprap size/weight relationships

Sieve Size	Approx. Weight of Rock	Weight of Test Pile
16 inches	300 pounds	6,000 pounds
11 inches	100 pounds	2,000 pounds
6 inches	15 pounds	300 pounds

- e. The results of the test must be compared to the gradation required for the project. Test pile results that do not meet the construction specifications must cause the rock to be rejected. The test pile that meets contract requirements must be left on the jobsite as a sample for visual comparison. The test pile must be used as part of the last rock riprap to be placed.
2. Filter or bedding aggregates, when required, must conform to Material Specification 521, Aggregates for Drainfill and Filters, unless otherwise specified.
3. Portland cement must conform to the requirements of Material Specification 531 for the specified type.
4. Pozzolan conforming to Specification ASTM C618, Class C or F, in amounts not to exceed 25 percent based on absolute volume, may be substituted for an equivalent amount of portland cement in the grout mixture unless otherwise specified in Section M of this specification.
5. Aggregates must conform to the requirements of Material Specification 522, Aggregates for Portland Cement Concrete, except that the grading for coarse aggregate must be as specified in Section M of this specification.
6. Water must be clean and free from detrimental amounts of oils, acid, alkali, organic matter, or other deleterious substances.
7. Air-entraining admixtures must conform to the requirements of Material Specification 533, Chemical Admixtures for Concrete.
8. Curing compound must conform to the requirements of Material Specification 534, Concrete Curing Compound.
9. Other admixtures, when required, must be as specified in Section M of this specification.
10. Geotextiles must conform to the requirements of Material Specification 592.

C. Subgrade Preparation

1. The subgrade surface on which the grouted rock riprap, filter, bedding, or geotextile is to be placed must be cut or filled and graded to the lines and grades shown on the drawings. When fill to subgrade lines is required, it must consist of approved material and must conform to the requirements of the specified class of earthfill.
2. Rock riprap, filter, bedding, or geotextile must not be placed until the foundation preparation is completed and the subgrade surface has been inspected and approved.

D. Placement of Rock Riprap

1. Method 1: Equipment-Placed Rock.
 - a. Place the rock riprap by equipment on the surface and to the depth specified. It must be installed to the full section thickness in one operation and in such a manner that avoids serious displacement of the underlying material. Deliver the rock for riprap and place in a manner that ensures the in-place riprap is

reasonably homogeneous with the larger rocks uniformly distributed and firmly in contact one to another with the smaller rocks and spalls filling the voids between the larger rocks. Some hand-placing may be required to provide a neat and uniform surface.

- b. Place rock riprap in a manner that prevents damage to structures. Hand-placing is required as necessary to prevent damage to any new and existing structures.
2. Method 2 Hand-Placed Rock. Place the rock riprap by hand on the surface and to the depth specified. It must be securely bedded with the larger rocks firmly in contact one to another without bridging. Fill spaces between the larger rocks with smaller rocks and spalls. Smaller rocks must not be grouped as a substitute for larger rock. Lay flat slab rock on its vertical edge (except where it is installed like paving stone). The thickness of the rock equals the specified depth of the riprap section.

E. Filter or Bedding

1. When the contract specifies filter, bedding, or geotextile beneath the rock riprap, place the designated material on the prepared subgrade surface as specified. Compaction of filter or bedding aggregate is not required, but finish the surface of such material reasonably smooth and free of mounds, dips, or windrows.

F. Design of the Grout Mix

1. The mix proportions for the grout mix as specified in the construction details in Section M of this specification. During installation, the engineer may require adjustment of the mix proportions whenever necessary. The mix must not be altered without the approval of the engineer.

G. Handling and Measurement of Grout Material

1. Material must be stockpiled and batched by methods that prevent segregation or contamination of aggregates and ensure accurate proportioning of the mix ingredients.
2. Except as otherwise provided in Section M of this specification, measure cement and aggregates as follows:
 - a. Measure cement by weight or in bags of 94 pounds each. When cement is measured in bags, no fraction of a bag may be used unless it is weighed.
 - b. Measure aggregates by weight. Base mix proportions on the batch weight of each aggregate saturated, surface-dry weight plus the weight of surface moisture it contains at the time of batching.
 - c. Measure water by volume or weight to an accuracy within 1 percent of the total quantity of water required for the batch.
 - d. Measure admixtures within a limit of accuracy of plus or minus 3 percent.

H. Mixers and Mixing

1. The mixer, when operating at capacity, must be capable of combining the ingredients of the grout mix into a thoroughly mixed and uniform mass and of discharging the mix with a satisfactory degree of uniformity.
2. Operate the mixer within the limits of the manufacturer's guaranteed capacity and speed of rotation.
3. The mixing time after all cement and aggregates have been combined in the mixer must be a minimum of 1 minute for mixers having a capacity of 1 cubic yard or less. For larger-capacity mixers, the minimum time must be increased 15 seconds for each cubic yard or fraction thereof of additional capacity. The batch must be so charged into the mixer that some water will enter in advance of the cement and aggregates, with the balance of the mixing water introduced into the mixer before a fourth of the total minimum mixing time has elapsed.
4. When ready-mix grout is furnished, the contractor must furnish to the engineer at the time of delivery a ticket showing the time of loading and the quantities of material used for each load of grout mix delivered.
5. No mixing water more than the amount required by the approved job mix must be added to the grout mix during mixing or hauling or after arrival at the delivery point.

I. Conveying and Placing

1. Deliver the grout mix to the site and place within 1.5 hours after the introduction of the cement to the aggregates. In hot weather or under conditions contributing to accelerated stiffening of the concrete, the time between the introduction of the cement to the aggregates and complete discharge of the grout batch must be a maximum of 45 minutes. The engineer may allow a longer time provided the setting time of the grout is increased a corresponding amount by the addition of an approved set-retarding admixture. In any case, concrete must be conveyed from the mixer to the final placement as rapidly as practicable by methods that prevent segregation of the aggregates, loss of mortar, displacement of the rock riprap, or a combination of these.
2. Grout mix must not be allowed to free fall more than 5 feet unless suitable equipment is used to prevent segregation.
3. The grout mix must not be placed until the rock riprap has been inspected and approved by the engineer for the placement of grout.
4. Rock to be grouted must be kept moist for a minimum of 2 hours before grouting.
5. The rock riprap must be flushed with water before placing the grout to remove fines from the rock surfaces. The rock must be kept moist before the grouting and without placing in standing or flowing water. Grout placed on inverts or other nearly level areas may be placed in one operation. On slopes, the grout must be placed in two nearly equal applications consisting of successive lateral strips about 10 feet in width starting at the toe of the slope and progressing upward.

Deliver the grout to the final place of deposit by approved methods and discharged directly on the surface of the rock. A metal or wood splash plate is used to prevent displacement of the rock directly under the grout discharge. Direct the grout flow with brooms, spades, or baffles to prevent grout from flowing excessively along the same path and to ensure that all intermittent spaces are filled. Sufficient barring must be conducted to loosen tight pockets of rock and otherwise aid in the penetration of grout to ensure the grout fully penetrates the total thickness of the rock blanket. All brooming on slopes must be uphill. After the grout has stiffened, the entire surface must be re-broomed to eliminate runs and to fill voids caused by sloughing. The surface finish, following the completion of grout installation, must consist of one third of the rock extended above the level of grout. The exposed rock will not have a plastered appearance.

6. After completion of any strip or panel, no individuals or equipment may be permitted on the grouted surface for 24 hours. Protect the grouted surface from sun, rain, flowing water, mechanical injury, or other potential damaging activity.

J. Curing and Protection

1. Prevent the completed finished surface from drying for a minimum curing period of 7 days following placement. Exposed surfaces must be maintained in a moist condition continuously for the 7-day curing period or until curing compound has been applied as specified in this section. Maintain moisture by sprinkling, flooding, fog spraying, or covering with continuously moistened canvas, cloth mats, straw, sand, or other approved material. Use water or moist covering to protect the grout during the curing process without causing damage to the grout surface by erosion or other mechanisms that may cause physical damage.
2. The grouted rock may be coated with an approved curing compound as an alternative method to maintaining continuous moisture during the curing period. Spray the compound on the moist grout surface as soon as free water has disappeared and all surface finishing has been completed. Apply the compound at a minimum uniform rate of 1 gallon per 175 square feet of surface and form a continuous, adherent membrane over the entire surface. Curing compound must not be applied to surfaces requiring bonding to subsequently placed grout and concrete. If the membrane is damaged during the curing period, the damaged area must be resprayed at the application rate specified for the original treatment.
3. Grout mix must not be placed when the daily minimum temperature is less than 40 degrees Fahrenheit unless facilities are provided to ensure that the temperature of the material is maintained at a minimum temperature of 50 degrees Fahrenheit and not more than 90 degrees Fahrenheit during placement and the curing period. Grout mix must not be placed on a frozen surface. When freezing conditions prevail, rock to be grouted must be covered and heated within a range of 50 to 90 degrees Fahrenheit for a minimum of 24 hours before placing the grouting material.

K. Inspecting and Testing Fresh Grout

1. Check and test the grout material throughout the grouting operation. Sampling fresh grout must be conducted in conformance with ASTM C172. The volume of each batch will be determined by methods prescribed in ASTM C138.
2. The engineer must have free access to all parts of the contractor's plant and equipment used for mixing and placing grout during the period of the contract. Proper facilities must be provided for the engineer to sample material and view processes implemented in the mixing and placing of grout as well as for securing grout test samples. All tests and inspections must be conducted so that only a minimum of interference to the contractor's operation occurs.
3. For ready-mixed grout, the contractor must furnish to the engineer a statement-of-delivery ticket for each batch delivered to the site. At a minimum, the ticket must provide weight in pounds of cement, aggregates (fine and coarse), water, weight in ounces of air-entraining agent, time of loading, and the revolution counter reading at the time batching was started.

L. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, the volume of grouted rock riprap, including filter layers or bedding, is determined to the nearest cubic yard from the specified thickness shown on the drawings and the area on which acceptable placement has been installed. Payment for grouted rock riprap is made at the contract unit price. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the grouted rock riprap, filter layers and bedding, and geotextile material.
2. Method 2. For items of work for which specific unit prices are established in the contract, the volume of riprap and the volume of filter layers or bedding are determined to the nearest cubic yard from the specified thickness shown on the drawings and the area in which acceptable placement has been installed. The volume of grout is determined from the calculated batch volume and the number of mixed batches delivered to the site and placed in accordance with the specification. The area of geotextile is determined to the nearest square yard from measurements of geotextile material installed according to the contract requirements. Payment is made at the contract unit price for each type of rock riprap, filter or bedding, concrete grout, and geotextile. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work.
3. Method 3. For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest 0.1 ton by actual weight. The quantity of each type of filter or bedding aggregate delivered and placed within the specified limits is computed to the nearest 0.1 ton. For each load of rock riprap placed as specified, the contractor must furnish to the engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton. For each load of filter or bedding aggregate, the

contractor must furnish to the engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton. The volume of grout is determined from the calculated batch volume and the number of mixed batches delivered to the site and placed in accordance with the specifications and drawings. The area of geotextile is determined to the nearest square yard from measurements of geotextile material installed according to the contract requirements. Payment is made at the contract unit price for each type of rock riprap, filter or bedding, concrete grout, and geotextile. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work.

4. All Methods. The following provision applies to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in Section M of this specification.

M. Items of Work and Construction Details

Items of work to be performed in accordance with this specification and details are:

1. General

- a. This item shall consist of furnishing and placing the grout and grouted rock riprap as shown on the drawings for the NF Baptist Church site.
- b. This item shall also include the removal and disposal of trash and debris within the work areas. Refuse material shall be hauled to an approved off-site landfill. The contractor shall be responsible for landfill tipping fees.

2. Bid Item 3A, Rock Riprap

- a. This item shall consist of furnishing and placing the rock riprap for the grouted rock riprap as shown on the drawings.
- b. This item shall also include the removal and disposal of trash and debris within the work areas. Refuse material shall be hauled to an approved off-site landfill. The contractor shall be responsible for landfill tipping fees.
- c. Section B, Material, rock shall meet the requirements of Rock Type 1, Material Specification 523, or be from sources with current approval from the WV DOT. The contractor shall provide a gradation test as specified with the material certification prior to delivery of rock riprap to the site.
- d. The statement in Section B, Material, that begins with “At least 30 days before rock is delivered ...” is to be changed to read “At least 7 days before rock is delivered...”.
- e. Method 1 of Section D, Placement of Rock Riprap shall apply. The final placement shall have enough voids to accommodate the concrete grout. Rock shall be hand-placed around existing structures, any existing pipes that need to be extended through the grouted rock riprap, and the new drainage pipes that are to extend through the grouted rock riprap.
- f. The rock riprap shall be well-graded between the nominal sizes of 6 and 18 inches. Well-graded shall mean that a broad range of rock particle sizes are represented and that substantial amount of each rock particle are included. Not more than 15 percent of the rock particles by weight shall be smaller than 6 inches. Not more than 50 percent of the rock particles by weight shall be smaller than 12 inches. One hundred percent of the rock particles shall be smaller than 22 inches.
- g. Method 3 of Section L, Measurement and Payment, shall apply for the Rock Riprap, except that the filter or bedding material is subsidiary to Bid Item 3A, Rock Riprap and will not be measured.
- h. Items of work subsidiary to this bid item include:
 - i. Channel Clearing and Shaping, Construction Specification 4
 - ii. Pollution Control, Construction Specification 5
 - iii. Traffic Control, Construction Specification 9
 - iv. Removal of Water, Construction Specification 11
 - v. Excavation, Construction Specification 21
 - vi. Earthfill, Construction Specification 23
 - vii. Drainfill, Construction Specification 24

- viii. Rockfill, Construction Specification 25
- ix. Steel Reinforcement, Construction Specification 34
- x. Plastic Pipe, Construction Specification 45
- xi. Contractor Quality Control, Construction Specification 94
- xii. Geotextile, Construction Specification 95

3. Bid Item 4A, Grout for Rock Riprap

- a. This item shall consist of furnishing and placing the grout for grouted rock riprap as shown on the drawings.
- b. The concrete grout mix shall be as follows:
 - i. Cement shall be Type I or IA.
 - ii. Minimum cement content shall be 6 bags per cubic yard or 564 pounds per cubic yard.
 - iii. The water-cement ratio shall not exceed 0.50.
 - iv. Coarse aggregate shall be ASTM C33 No. 57 or smaller.
 - v. Fine aggregate shall be ASTM C33 Fine Aggregate.
 - vi. Air content shall be 4 to 8 percent.
 - vii. Ratio of fine to coarse aggregate shall be as recommended by the supplier.
 - viii. Superplasticizer shall be used in the grout mix to improve workability.
 - ix. Curing compound shall be used for grout curing.
- c. The slump for the grout prior to superplasticizer shall be a maximum of 3 inches. The maximum slump after adding superplasticizer and prior to placement shall be 7-1/2 inches. After obtaining slump tests on-site that verifies the 3 inch or less slump, the contractor shall add a Type D (Plasticizing and Retarding) admixture which meets the requirements of ASTM C 494 and Material Specification 533-Chemical Admixtures for Concrete. The amount of Type D admixture shall be per the manufacturer's recommendations which will result in a slump, at point of placement, of 6 to 7-1/2 inches, with no additional water added. The grout mixture shall be such that the grout may be readily placed without segregation of materials or excessive laitance.
- d. Curing compound shall comply with ASTM C 1315, Type I, Class A.
- e. The grout shall be consolidated into the voids with the use of a concrete vibrator. The concrete vibrator shall be an internal type mechanical vibrator capable of transmitting vibration to the concrete at frequencies not less than 8,000 impulses per minute, unless otherwise specified or approved before placement. The construction contractor shall have two vibrators available on-site in the event one vibrator fails during placement. A smooth surface is not to be created during the grouting operation.
- f. The concrete grouting operation shall be performed in the presence of the Engineer or the Engineer's representative.
- g. Grouted Rock Riprap placement during cold weather shall be as described in Section J, Curing and Protection, of this specification. The curing period during which concrete temperature shall be maintained at a minimum of 50

degrees Fahrenheit shall be 7 days. Concrete surface temperature shall be checked daily with a max/min thermometer, and the temperatures shall be recorded for and provided to the Engineer.

- h. Method 2 of Section L, Measurement and Payment, shall apply for the Grout.

Construction Specification 64 — Gabions and Gabion Mattresses

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing, assembling, and installing rock-filled wire-mesh gabions and gabion mattresses. Gabions are at least 12 inches high. Gabion mattresses are no more than 12 inches high.

B. Material Types

1. Gabions and gabion mattresses consist of rectangular wire mesh formed containers filled with rock. Gabions must conform to one of the following types. The wire coating must be as specified in section G.
 - a. Twisted-mesh.
 - (1) Nonraveling, double twisted, hexagonal wire mesh consisting of two wires twisted together in two 180-degree turns. Twisted-mesh, fasteners, and stiffeners must conform to the requirements of ASTM A975.
 - (2) Lacing wire is the standard fastener for twisted-mesh gabions and gabion mattresses. Ring fasteners may be used and must be made of stainless steel. Ring fasteners must provide the minimum strength per lineal foot that is specified in ASTM A975 for gabions and gabion mattresses.
 - b. Welded mesh.
 - (1) Welded mesh with a uniform square or rectangular pattern and a resistance weld at each intersection. Welded mesh and stiffeners must conform to the requirements of ASTM A974 with the exception that welded-mesh may be delivered in component form, either rolled or stacked, for assembly at the jobsite.
 - (2) Spiral binders are the standard fastener for welded-mesh gabions and gabion mattresses. Spiral binders must conform to the requirements of ASTM A974. Alternate fasteners for use with welded-mesh gabions and gabion mattresses, such as ring fasteners or lacing wire, must be formed from wire meeting the same quality and coating thickness requirements as specified for the gabions and gabion mattresses. Ring fasteners must be made of stainless steel. Standard fasteners and alternate fasteners must provide the minimum strength per lineal foot that is specified in section G.
2. Rock must conform to the quality requirements in Material Specification 523, Rock for Riprap, unless otherwise specified in Section G. At least 85 percent of

the rock particles (by weight) must be within the predominant rock size range shown in figure 64.1.

Figure 64.1 - Predominant Rock Size Range by Gabion Size

Gabion Basket or Mattress Height	Predominant Rock size (in)	Minimum Rock Dimension (in)	Maximum Rock Dimension (in)
12-, 18-, or 36-inch basket	4 to 8	4	8
6-, 9-, or 12-inch mattress	3 to 6	3	6

3. At least 30 days before delivery to the site, the contractor must inform the engineer in writing of the source from which the rock will be obtained. The test data and other information by which the material was determined by the contractor to meet the requirements of this specification must be included. The contractor must provide the engineer free access to the source for the purpose of obtaining samples for testing and source approval.
4. Bedding or filter material, when specified, must meet the gradation shown on the plans or as specified in section G. The requirements of Material Specification 521, Aggregates for Drainfill and Filters. Geotextile, when specified, must conform to the requirements specified in section G and in Material Specification 592, Geotextile.

C. Foundation Preparation

1. The foundation on which the gabions and gabion mattresses are to be placed must be cut or filled and graded to the lines and grades shown on the drawings. Surface irregularities, loose material, vegetation, and all foreign matter must be removed from the foundation. When fill is required, it must consist of material conforming to the requirements specified in section G. Fill must be compacted as specified in section G. Gabions, gabion mattresses, and bedding or specified geotextiles may not be placed until the foundation preparation is completed, inspected by the responsible engineer, and verified by the engineer to meet the requirements of this specification.
2. Compaction of bedding or filter material is required as specified in section G. The surface of the finished material must be to grade and free of mounds, dips, or windrows. Geotextile must be installed in accordance with the requirements of Construction Specification 95.

D. Assembly and Placement

1. Where a gabion or gabion mattress unit must be modified, welded-mesh panels may be cut to fit. Twisted-mesh panels may be folded and neatly wired but must not be cut.
2. Assembly.
 - a. Use ring-type fasteners or lacing wire for the assembly and placement of twisted-mesh panels. Wrap the lacing wire with alternating single and double

half-hitches at 4- to 6-inch intervals and secure by tying a double half-hitch at each end. Use spiral or ring type fasteners for the assembly and placement of welded-mesh panels. Where spiral fasteners are used, crimp the ends to secure the spirals in place. Where ring type fasteners are used, install the fasteners at a maximum spacing of 6 inches.

- b. Interior diaphragms are required where any inside dimension exceeds 3 feet. Use the same type fasteners and fastening procedures to install interior diaphragms and lids as used in the panel assembly. Diaphragms are installed to ensure that no open intervals are present that exceed 3 feet.

3. Placement.

- a. Place the empty gabions or gabion mattresses on the foundation and use lacing wire to connect all adjacent gabions along the top, bottom, and vertical edges. Wrap the wire with alternating single and double half-hitches at 4- to 6-inch intervals. Unless otherwise specified in section G, lacing wire is the only fastener allowed for connecting twisted-mesh gabions. Welded-mesh gabions and gabion mattresses may be connected with spiral fasteners, ring-type fasteners, or lacing wire. Where spiral fasteners are used, crimp the ends to secure the spirals in place. Where ring-type fasteners are used, install the fasteners at a maximum spacing of 6 inches. Lacing wire may be used as needed to supplement the interconnection of welded mesh gabions and the closing of lids.
- b. Connect each layer of gabions and gabion mattresses to the underlying layer of gabions along the front, back, and sides. Stagger the vertical and horizontal joints between the gabions of adjacent rows and layers by at least one-fourth of a cell length.

E. Filling Operation

1. Twisted-mesh.

- a. After adjacent empty twisted-mesh units are set to line and grade and common sides properly connected, they must be placed in straight line and stretched to remove any kinks from the mesh and gain a uniform alignment. Units may be staked to maintain the established proper alignment before the rock is placed. No stakes may be placed through geotextile material. Fasteners must be attached during the filling operation as needed to preserve the strength and shape of the structure.
- b. Internal connecting crosstie wires must be placed in each unrestrained gabion and gabion mattress unit of more than 18 inches in height, including units left temporarily unrestrained. Two internal connecting wires must be placed concurrently with rock placement at each 12-inch interval of depth. These crossties must be evenly spaced along the front face and connected to the back face. All crosstie wires must be looped around two mesh openings and each wire end must be secured by a minimum of five 180-degree twists around itself after looping.

2. Welded-mesh.

- a. Welded-mesh units do not require stretching. Units may be staked to maintain the established proper alignment before the rock is placed. No stakes may be placed through geotextile material. Fasteners must be attached during the filling operation as needed to preserve the strength and shape of the structure.
 - b. Internal crossties or stiffeners must be placed in each unrestrained gabion and gabion mattress unit of more than 18 inches in height, including units left temporarily unrestrained. Crossties or stiffeners must be placed concurrently with rock placement at each 12-inch interval of depth. They must be placed across the corners of the gabions (at 12 inches from the corners) providing diagonal bracing. Lacing wire or preformed hooked wire stiffeners may be used.
3. Twisted and welded-mesh.
 - a. The gabions and gabion mattresses must be carefully filled with rock in a manner that will ensure alignment, avoid bulges, and provide a compact mass that minimizes voids. Machine placement requires supplementing with hand work to ensure the desired results. The units or cells in any row must be filled in stages so that the depth of rock placed in any one cell does not exceed the depth of rock in any adjoining cell by more than 12 inches. Along the exposed faces, the outer layer of stone must be carefully placed and arranged by hand to ensure a neat, compact placement with a uniform appearance.
 - b. The last layer of rock must be uniformly leveled to the top edges of the cells. Lids must be stretched tight over the rock filling. The use of crowbars or other single point leverage bars for lid closing is prohibited as they may damage the baskets. The lid must be stretched until it meets the perimeter edges of the front and end panels. The gabion lid must then be secured to the sides, ends, and diaphragms with lacing wire, spiral binders, or approved alternate fasteners. Lacing wire must be wrapped with alternating single and double half-hitches at 4- to 6-inch intervals. Where spiral fasteners are used, crimp the ends to secure the spirals in place.
4. All Styles. Any damage to the wire or coatings during assembly, placement, and or filling must be repaired promptly in accordance with the manufacturer's recommendations or replaced with undamaged gabion basket materials.

F. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, the volume of rock is measured within the neat lines of the gabion structure and computed to the nearest cubic yard. Payment for gabions is made at the contract unit price and includes the wire mesh and rock. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to completion of the work.
2. Method 2. For items of work for which specific unit prices are established in the contract, the volume of the gabions is measured within the neat lines of the gabion structure and computed to the nearest cubic yard. Payment for the gabions is made at the contract unit price and includes the wire mesh, rock, and specified bedding material or geotextile. Such payment is considered full compensation for all labor,

material, equipment, and other items necessary and incidental to the completion of the work.

3. Method 3. For items of work for which specific unit prices are established in the contract, the surface area is measured within the neat lines of the gabion mattress structure and computed to the nearest square yard. Payment for the gabion mattress is made at the contract unit price and includes the wire mesh and rock. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work.
4. Method 4. For items of work for which specific unit prices are established in the contract, the surface area is measured within the neat lines of the gabion mattress structure and computed to the nearest square yard. Payment for the gabion mattress is made at the contract unit price and includes the wire mesh, rock, and specified bedding material or geotextile. Such payment is considered full compensation for all labor, material, equipment, and other items necessary and incidental to the completion of the work.
5. All Methods. The following provisions apply to all methods of measurement and payment. Unless otherwise specified in section G, no deduction in volume is made for any void or embedded item (e.g., a pipe passing through a gabion wall). Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Each item and the items to which they are made subsidiary are identified in section G of this specification.

G. Items of Work and Construction Details

Items of work to be performed according to this specification and the construction details are:

1. Bid Item 5B, Gabions and Gabion Mattresses, Twisted-Mesh Gabions, Site 2.

- a. This item shall consist of furnishing, assembling, and installing rock-filled wire mesh gabions as shown on the drawings or staked in the field. This item shall also include all excavation necessary to install the gabion wall protection to neat lines and grades.
- b. Included in this item is the removal and disposal of trash and debris within the work area. Refuse material shall be hauled to an approved off-site landfill. Contractor shall be responsible for landfill tipping fees.
- c. In Section B, Material Types, gabions shall be twisted-mesh baskets. The gabion baskets shall also include diaphragms.
- d. In Section B, Material Types:
 - i. Gabion basket wire and all other wire shall be coated with Gray PVC. Twisted-wire mesh shall be zinc-coated and coated with PVC (ASTM A975 Style 3).
 - ii. All fastenings shall be made by lacing for twisted-mesh. Ring fasteners or other alternative fastening methods are prohibited.
 - iii. Rock shall conform to the requirements of Material Specification 523, Type 1 or be from a source with current approval with the WV DOH.
 - iv. In Section B, “30 days” is changed to “7 days”.
- e. In Section D, Assembly and Placement, twisted-mesh baskets shall be assembled using lacing wire and welded-mesh baskets shall be assembled using spiral binders.
- f. In Section E, Filling Operation, hand placement is required to assure no rocks smaller than 4 inches are exposed on visible faces. The rock on visible faces are also required to be placed such that flat edges are turned out to make a smooth surface. Hand placing is also required to remove voids around all edges of baskets and under the crosstie wires.
- g. Gabion baskets are to be unfolded and all kinks and unwanted folders are to be removed prior to assembling.
- h. Gabion baskets may be assembled off-site. Off-site is defined as work areas where work is being performed at the contractor’s headquarters, tool yards, etc. and must not be dedicated to the exclusive performance of the federal contract.
- i. Gabions will be backfilled with coarse-grain materials obtained from the required excavations. Where adequate material is not present, the

contractor shall obtain a suitable supply from an off-site location. All backfill materials will be approved by the NRCS Engineer or Inspector before placement.

- j. In Section F, Measurement and Payment, Method 2 shall apply.
- k. Items of work subsidiary to this bid item are:
 - i. Channel Clearing and Shaping, Construction Specification 4
 - ii. Pollution Control, Construction Specification 5
 - iii. Traffic Control, Construction Specification 9
 - iv. Removal of Water, Construction Specification 11
 - v. Excavation, Construction Specification 21
 - vi. Earthfill, Construction Specification 23
 - vii. Drainfill, Construction Specification 24
 - viii. Rockfill, Construction Specification 25
 - ix. Rock Riprap, Construction Specification 61
 - x. Contractor Quality Control, Construction Specification 94
 - xi. Geotextile, Construction Specification 95

Construction Specification 94 — Contractor Quality Control

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of developing, implementing, and maintaining a quality control system to ensure that the specified quality is achieved for all materials and work performed.

B. Equipment and Materials

1. Equipment and material used for quality control must be of the quality and condition required to meet the test specifications cited in the contract. Testing equipment must be properly adjusted and calibrated at the start of operations with the calibration maintained at the frequency specified. Records of equipment calibration tests must be available to the engineer at all times. Equipment must be operated and maintained by qualified operators as prescribed in the manufacturer's operating instructions, the references specified, and as specified in Section J of this specification. All equipment and materials used in performing quality control testing must be as prescribed by the test standards referenced in the contract or in Section J.
2. Handle and operate all equipment and materials in a safe and proper manner and comply with all applicable regulations pertaining to their use, operation, handling, storage, and transportation.

C. Quality Control System

1. Method 1. The contractor must develop, implement, and maintain a system of quality control to provide the specified material testing and verification of material quality before use. The system activities must include procedures to verify adequacy of completed work, initiate corrective action to be taken, and document the final results. The identification of the quality control personnel and their duties and authorities must be submitted to the contracting officer in writing within 15 calendar days after notice of award.
2. Method 2. The contractor must develop, implement, and maintain a system adequate to achieve the specified quality of all work performed, material incorporated, and equipment furnished before use. The system established must be documented in a written plan developed by the contractor and approved by the contracting officer. Include in the system activities the material testing and inspection needed to verify the adequacy of completed work and procedures to be followed when corrective action is required. Daily records to substantiate the

- conduct of the system must be maintained by the contractor. The quality control plan must cover all aspects of quality control and must address, at a minimum, all specified testing and inspection requirements. Provide the plan consistent with the planned performance in the contractor's approved construction schedule. Identify in the plan the contractor's onsite quality control manager and provide an organizational listing of all quality control personnel and their specific duties. The written plan must be submitted to the contracting officer within 15 calendar days after notice of award. The contractor must not proceed with any construction activity that requires inspection until the written plan is approved by the contracting officer.
3. All Methods. The quality control system must include, but not be limited to, a rigorous examination of construction material, processes, and operation, including testing of material and examination of manufacturer's certifications as required, to verify that work meets contract requirements and is performed competently.

D. Quality Control Personnel

1. Method 1. Quality control activities must be accomplished by competent personnel. A competent person is one who is experienced and capable of identifying, evaluating, and documenting that materials and processes being used will result in work that complies with the contract. This person also has authority to take prompt action to remove, replace, or correct such work or products not in compliance. Off-site testing laboratories must be certified or inspected by a nationally recognized entity. The contractor must submit to the contracting officer for approval laboratory certification or inspection information. The contractor must submit to the contracting officer for approval the names, qualifications, authorities, certifications, and availability of the competent personnel who will perform the quality control activities.
2. Method 2. Quality control activities must be accomplished by competent personnel who are separate and apart from line supervision and report directly to management. A competent person is experienced and capable of identifying, evaluating, and documenting that material and processes being used will result in work that complies with the contract. This person has authorization to take prompt action to remove, replace, or correct such work or products not in compliance. Off-site testing laboratories must be certified or inspected by a nationally recognized entity. The contractor must submit to the contracting officer for approval laboratory certification or inspection information. The contractor must submit to the contracting officer for approval the names, qualifications, authorities, certifications, and availability of the competent personnel who will perform the quality control activities.

E. Post-award Conference

1. The contractor must meet with the contracting officer before any work begins and discuss the contractor's quality control system. The contracting officer and the contractor must develop a mutual understanding regarding the quality control system including procedures for correcting quality control issues.

F. Records

1. The contractor's quality control records must document both acceptable and deficient features of the work and corrective actions taken. All records must be on forms approved by the contracting officer, be legible, and be dated and signed by the competent person creating the record.
2. Unless otherwise specified in Section J of this specification, records must include:
 - a. Documentation of shop drawings including date submitted to and date approved by the contracting officer, results of examinations, need for changes or modifications (if any), manufacturer's recommendations and certifications (if any), and signature of the authorized examiner.
 - b. Documentation of material delivered including quantity, storage location, and results of quality control examinations and tests.
 - c. Type, number, date, time, and name of individual performing quality control activities.
 - d. The material or item inspected and tested, the location and extent of such material or item, and a description of conditions observed and test results obtained during the quality control activity.
 - e. The determination that the material or item met the contract provisions and documentation that the engineer was notified.
 - f. For deficient work, the nature of the defects, specifications not met, corrective action taken, and results of quality control activities on the corrected material or item.

G. Reporting Results

1. The results of contractor's quality control inspections and tests must be communicated to the engineer immediately upon completion of the inspection or test. Unless otherwise specified in Section J, the original plus one copy of all records, inspections, tests performed, and material testing reports must be submitted to the engineer within 1 working day of completion. The original plus one copy of documentation of material delivered must be submitted to the engineer before the material is used.

H. Access

1. The contracting officer and the engineer must be given free access to all testing equipment, facilities, sites, and related records for the duration of the contract.

I. Payment

1. Method 1.
 - a. For items of work for which lump sum prices are established in the contract, payment is made as the work proceeds after presentation by the contractor of invoices showing related costs and evidence of charges by suppliers, subcontractors, and others for furnishing supplies and work performed. If the total of such payments is less than the lump sum contract price for this item, the remaining balance is included in the final contract payment. Payment of

the lump sum contract price constitutes full compensation for completion of the work.

- b. Payment is not made under this item for the purchase cost of material and equipment having a residual value.

2. Method 2.

- a. For items of work for which lump sum prices are established in the contract, payment is prorated and paid in equal amounts on each monthly estimate. The number of months used for prorating must be the number estimated to complete the work. The final month's prorate amount is made with the final payment. Payment as described above constitutes full compensation for completion of the work.
- b. Payment is not made under this item for the purchase cost of material and equipment having a residual value.

3. All Methods. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section J.

J. Items of work and construction details

Items of work to be performed in conformance with this specification and construction details are:

a. Subsidiary Item, Contractor Quality Control

- (1) This item shall consist of all work and activities necessary to ensure work meets the specified lines, grades, and quality for the work included in this contract.
- (2) The contractor shall be required to perform slump, air (volumetric), and temperature tests for each batch of concrete mixed. A slump test shall be performed both before and after the addition of the plasticizing admixture.
- (3) In Section C, Quality Control System, Method 1 shall apply. Quality control personnel will be identified within 2 calendar days after notice to proceed.
- (4) In Section D, Quality Control Personnel, Method 1 shall apply.
- (5) In Section I, Payment, shall not apply. No separate payment will be made for Contractor Quality Control. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Construction Specification 95 — Geotextile

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. This work consists of furnishing all material, equipment, and labor necessary for the installation of geotextiles.

B. Quality

1. Geotextiles must conform to the requirements of Material Specification 592 and this specification.

C. Storage

1. Before use, store the geotextile in a clean, dry location out of direct sunlight, not subject to extremes of either hot or cold temperatures, and with the manufacturer's protective cover undisturbed. Receiving, storage, and handling at the jobsite must be in accordance with the requirements listed in ASTM D4873.

D. Surface preparation

1. The surface on which the geotextile is to be placed must be graded to the neat lines and grades as shown on the drawings. The surface must be reasonably smooth and free of loose rock and clods, holes, depressions, projections, muddy conditions, and standing or flowing water (unless otherwise specified in section G of this specification).

E. Placement

1. Before the geotextile is placed, the soil surface will be reviewed for quality assurance of the design and construction. Place the geotextile on the approved prepared surface at the locations and in accordance with the details shown on the drawings and specified in section G of this specification. It must be unrolled along the placement area and loosely laid, without stretching, in such a manner that it conforms to the surface irregularities when material or gabions are placed on or against it. The geotextile may be folded and overlapped to permit proper placement in designated areas.
2. Method 1.
 - a. Join the geotextile by machine sewing using thread material meeting the chemical requirements for the geotextile fibers or yarn. Thread must be polypropylene, polyester, or Kevlar™ aramid thread unless a specific thread type is specified. The thread must consist of two parallel stitched rows at a

spacing of about 1 inch and must not cross (except for any required re-stitching). The stitching must be a lock-type stitch. Each row of stitching must be located a minimum of 2 inches from the geotextile edge. Unless otherwise specified, the seam tensile strength as measured according to ASTM D4884 must be a minimum of 90 percent of the geotextile tensile strength in the weakest principal direction as measured according to ASTM D4632.

- b. The geotextile must be temporarily secured during placement of overlying material to prevent slippage, folding, wrinkling, or other displacement of the geotextile. Unless otherwise specified, methods of securing must not cause punctures, tears, or other openings to be formed in the geotextile.
3. Method 2.
 - a. Join the geotextile by overlapping a minimum of 18 inches (unless otherwise specified) and secured against the underlying foundation material. Securing pins, approved and provided by the geotextile manufacturer, must be placed along the edge of the panel or roll material to adequately hold it in place during installation. Pins must be steel or fiberglass formed as a U, L, or T shape or must contain ears to prevent total penetration through the geotextile. Provide steel washers on all but the U-shaped pins. The upstream or upslope geotextile must overlap the abutting downslope geotextile. At vertical laps, securing pins must be inserted through the bottom layers along a line through approximately the midpoint of the overlap. At horizontal laps and across slope laps, securing must be inserted through the bottom layer only. Securing pins must be placed along a line about 2 inches in from the edge of the placed geotextile at intervals not to exceed 12 feet unless otherwise specified. Install additional pins as necessary and where appropriate to prevent any undue slippage or movement of the geotextile. The use of securing pins will be held to the minimum necessary. Pins are to remain in place unless otherwise specified.
 - b. Should the geotextile be torn or punctured or the overlaps or sewn joint disturbed as evidenced by visible geotextile damage, subgrade pumping, intrusion, or grade distortion, the backfill around the damaged or displaced area must be removed and restored to the original approved condition. The repair must consist of a patch of the same type of geotextile being used and overlaying the existing geotextile. When the geotextile seams are required to be sewn, extend the overlay patch a minimum of 1 foot beyond the edge of any damaged area and join by sewing as required for the original geotextile except that the sewing must be a minimum of 6 inches from the edge of the damaged geotextile. Geotextile panels joined by overlap must have the patch extend a minimum of 2 feet from the edge of any damaged area.
4. Place geotextile in accordance with the following applicable specification according to the use indicated in section G:
 - a. Slope protection. The geotextile must not be placed until it can be anchored and protected with the specified covering within 48 hours or protected from exposure to ultraviolet light. In no case may material be dropped on uncovered geotextile from a height of more than 3 feet.

- b. Subsurface drains. The geotextile must not be placed until drainfill or other material can be used to provide cover within the same working day. Drainfill material must be placed in a manner that prevents damage to the geotextile. In no case may material be dropped on uncovered geotextile from a height of more than 5 feet.
- c. Road stabilization. The geotextile must be unrolled in a direction parallel to the roadway centerline in a loose manner permitting conformation to the surface irregularities when the roadway fill material is placed on its surface. In no case may material be dropped on uncovered geotextile from a height of more than 5 feet. Unless otherwise specified, the minimum overlap of geotextile panels joined without sewing must be 24 inches. The geotextile may be temporarily secured with pins recommended or provided by the manufacturer, but they must be removed before the permanent covering material is placed.

F. Measurement and payment

- 1. Method 1. For items of work for which specific unit prices are established in the contract, the quantity of geotextile for each type placed within the specified limits is determined to the nearest specified unit by measurements of the covered surfaces only, disregarding that required for anchorage, seams, and overlaps. Payment is made at the contract unit price. Such payment constitutes full compensation for the completion of the work.
- 2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of geotextile for each type placed within the specified limits is determined to the nearest specified unit by computing the area of the actual roll size or partial roll size installed. The computed area will include the amount required for overlap, seams, and anchorage as specified. Payment is made at the contract unit price. Such payment constitutes full compensation for the completion of the work.
- 3. Method 3. For items of work for which specific lump sum prices are established in the contract, the quantity of geotextile is not measured for payment. Payment for geotextiles is made at the contract lump sum price and constitutes full compensation for the completion of the work.
- 4. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section G of this specification.

G. Items of work and construction details

Items of work to be performed according to this specification and the construction details are:

1. Subsidiary Item, Geotextile

- a. This item shall consist of furnishing all labor, equipment, and materials required to install the geotextile as shown on the drawings.
- b. The geotextile shall be Class I, non-woven, and shall be supplied according to Material Specification 592. The apparent opening size (AOS) shall range from a maximum of #40 to a minimum of #80 (U.S. standard sieve size according to ASTM D4751).
- c. Place cover material over the geotextile in a manner that prevents damage to the geotextile. Geotextile damaged during placement or by equipment crossovers shall be replaced or repaired at no additional cost to the Government. All repairs shall be by replacement or by placing an additional sheet of geotextile over the damaged area, ensuring a minimum of 1 foot overlap on undamaged geotextile. Place geotextile so that no edge of the fabric are exposed outside the limits of the area to be covered.
- d. Method 2 of Section E, Placement, shall apply.
- e. In Section F, Measurement and Payment, shall not apply. No separate payment shall be made for Geotextile. Compensation for this item is subsidiary to and included in the payment for **Bid Item 3A, Rock Riprap, Construction Specification 62, Site 1 and Bid Item 5B, Gabions, Construction Specification 64, Site 2.**

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 521—Aggregates for Drainfill and Filters

A. Scope

This specification covers the quality of mineral aggregates for the construction of drainfill and filters.

B. Quality

- (1) Drainfill and filter aggregates shall be sand, gravel, or crushed stone or mixtures thereof. Aggregates shall be composed of clean, hard, durable, mineral particles free from organic matter, clay balls, soft particles, or other substances that would interfere with the free-draining properties of the aggregates.
- (2) Coarse aggregate may be crushed limestone or other material that has limestone particles included. Aggregates from crushed limestone shall be thoroughly washed and screened to remove limestone dust, limestone fines, and fine soil particles. Limestone shall not be used for fine aggregates except in combination with other material, such that not more than 5 percent of the portion finer than the No. 4 sieve shall be limestone.
- (3) Aggregates shall be tested for soundness according to ASTM C88 and shall have a weighted average loss in 5 cycles of not more than 12 percent when sodium sulfate is used or 18 percent when magnesium sulfate is used.

C. Grading

Drainfill and filter aggregates shall conform to the specified grading limits after being placed or after being compacted when compaction is specified. Grading shall be determined by ASTM C136. The percentage of material finer than the No. 200 sieve shall be determined by the method in ASTM C117.

D. Storing and Handling

Drainfill and filter aggregates shall be stored and handled by methods that prevent segregation of particle sizes or contamination by mixing with other material.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 522—Aggregates for Portland Cement Concrete

A. Scope

This specification covers the quality of fine aggregate and coarse aggregate for use in the manufacture of Portland cement concrete.

B. Quality

Aggregate shall conform to the requirements of ASTM C33 for the specified sizes. Aggregates that fail to meet any requirement may be accepted only when either:

- (i) The specified alternate conditions of acceptance can be proven before the aggregates are used on the job and within a period such that no work under the contract will be delayed by the requirements of such proof or,
- (ii) The specification for concrete expressly contains a provision of special mix requirements to compensate for the effects of the deficiencies.

C. Reactivity with alkalis

- (1) The potential reactivity of aggregates with the alkalis in cement shall be evaluated by petrographic examination as per ASTM C295, or by the results of previous tests or service records of concrete made from similar aggregates from the same source. The standards for evaluating potential reactivity shall be as described in ASTM C1778
- (2) Aggregates indicated by any of the above to be potentially reactive shall not be used except under one of the following conditions:
 - (i) Applicable test results of mortar bar tests made according to ASTM Method C 1567 are available which indicate an expansion of less than 0.10 percent at 16 days.
 - (ii) The concrete mixture complies with the appropriate testing procedures and mitigations measures established in ASTM C1778.
 - (iii) Concrete made from similar aggregates from the same source has been demonstrated to be sound after 3 years or more of service under conditions of exposure to moisture and weather similar to those anticipated for the concrete under these specifications.
- (3) Aggregates indicated to be potentially reactive, but within acceptable limits as determined by mortar bar test results or service records, shall be used only with low alkali cement, containing less than 0.60 percent alkalis expressed as sodium oxide.

D. Sulfur in Aggregate

- (1) There is currently not an ASTM standard on the acceptable level of sulfur in concrete aggregate.
- (2) To prevent concrete cracking from iron sulfide expansion, perform petrographic testing per ASTM C295 and use the following limits for sulfur (S) in aggregate based on the American Concrete Institute Technical Paper 113-M31 and the Concrete Society BS EN 12620:
 - (i) When S is less than 0.1 percent, the aggregate is acceptable.
 - (ii) When S is between 0.1 and 1.0 percent, perform further testing to determine if iron sulfide minerals such as pyrrhotite, gypsum, pyrite, or marcasite are present:
 - If the additional testing shows pyrrhotite, gypsum, pyrite, or marcasite are present, reject the aggregate and use an acceptable aggregate.

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- If the additional testing shows pyrrhotite, gypsum, pyrite, or marcasite are not present, the aggregate is acceptable.
- (iii) Aggregate with S greater than 1 percent is not acceptable.

E. Storing and Handling

Aggregates of each class and size shall be stored and handled by methods that prevent segregation of particles sizes or contamination by intermixing with other material.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 523—Rock for Riprap

A. Scope

This specification covers the quality of rock to be used in the construction of rock riprap.

B. Quality

- (1) Individual rock fragments must be dense, sound, and free from cracks, seams, and other defects conducive to accelerated weathering. Except as otherwise specified, the rock fragments must be angular to subrounded. The least dimension of an individual rock fragment must be not less than one-third the greatest dimension of the fragment. ASTM D4992 provides guidance on selecting rock from a source.
- (2) Except as otherwise provided, the rock must be tested and must have the following properties:
 - (i) Rock type 1
 - Bulk specific gravity (saturated surface-dry basis)—Not less than 2.5 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Absorption—Not more than 2 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Soundness—The weight loss in 5 cycles must not be more than 10 percent when sodium sulfate is used or more than 15 percent when magnesium sulfate is used.
 - (ii) Rock type 2
 - Bulk specific gravity (saturated surface-dry basis)—Not less than 2.5 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Absorption—Not more than 2 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Soundness—The weight loss in 5 cycles must be not more than 20 percent when sodium sulfate is used or more than 25 percent when magnesium sulfate is used.
 - (iii) Rock type 3
 - Bulk specific gravity (saturated surface-dry basis)—Not less than 2.3 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Absorption—Not more than 4 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
 - Soundness—The weight loss in 5 cycles must be not more than 20 percent when sodium sulfate is used or more than 25 percent when magnesium sulfate is used.

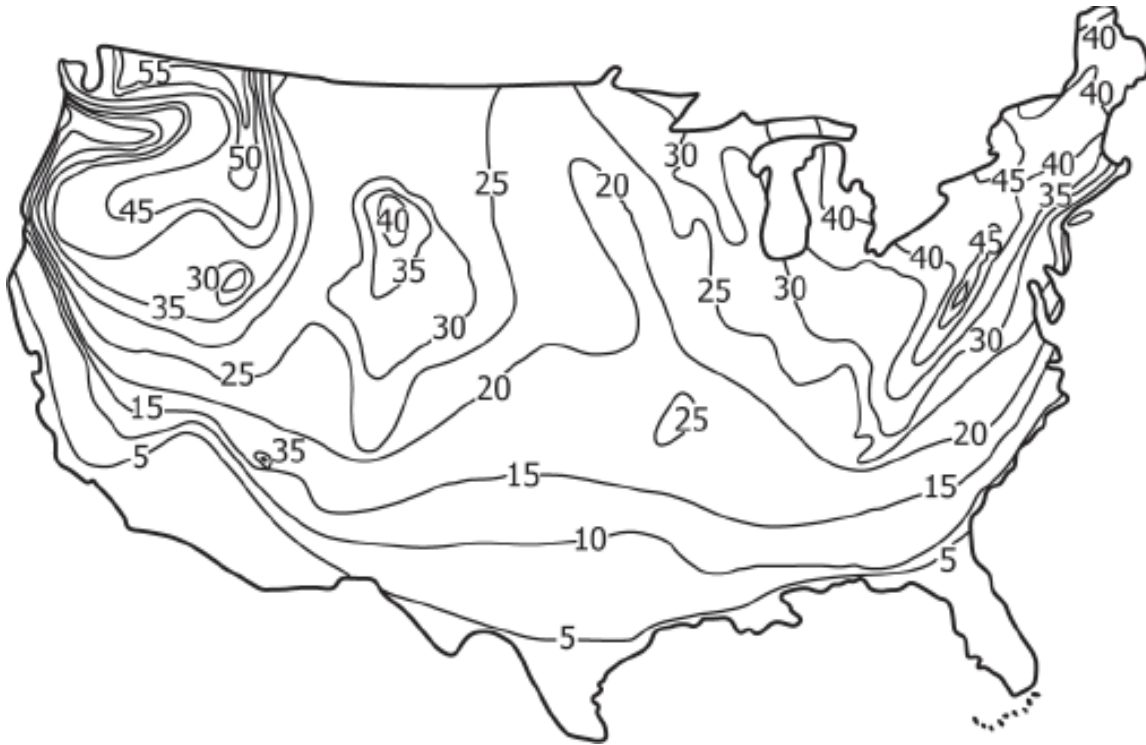
C. Methods of Soundness Testing

- (1) Rock cube soundness—
 - (i) The sodium or magnesium sulfate soundness test for all rock types (1, 2, or 3) must be performed on a test sample of $5,000 \pm 300$ grams of rock fragments, reasonably uniform in size and cubical in shape, and weighing, after sampling, about 100 grams each. They must be obtained from rock samples that are representative of the total rock mass, as noted in ASTM D4992, and that have been sawed into slabs as described in ASTM

D5121. The samples must further be reduced in size by sawing the slabs into cubical blocks. The thickness of the slabs and the size of the sawed fragments must be determined by the size of the available test apparatus and as necessary to provide, after sawing, the approximate 100-gram samples. The cubes must undergo five cycles of soundness testing in accordance with ASTM C88.

- (ii) Internal defects may cause some of the cubes to break during the sawing process or during the initial soaking period. Do not test any of the cubes that break during this preparatory process. Such breakage, including an approximation of the percentage of cubes that break, must be noted in the test report.
 - (iii) After the sample has been dried following completion of the final test cycle and washed to remove the sodium sulfate or magnesium sulfate, the loss of weight must be determined by subtracting from the original weight of the sample the final weight of all fragments that have not broken into three or more fragments.
 - (iv) The test report must show the percentage loss of the weight and the results of the qualitative examination.
- (2) Rock slab soundness—
- (i) When specified, the rock must also be tested in accordance with ASTM D5240. Deterioration of more than 25 percent of the number of blocks must be cause for rejection of rock from this source. Rock must also meet the requirements for average percent weight loss stated below.
 - (ii) For projects located north of the Number 20 Freeze-Thaw Severity Index Isoline (fig. 523–1 below), unless otherwise specified, the average percent weight loss for Rock Type 1 must not exceed 20 percent when sodium sulfate is used or 25 percent when magnesium sulfate is used. For Rock Types 2 and 3, the average percent weight loss must not exceed 25 percent for sodium sulfate soundness or 30 percent for magnesium sulfate soundness.
 - (iii) For projects located south of the Number 20 Freeze-Thaw Severity Index Isoline, unless otherwise specified, the average percent weight loss for Rock Type 1 must not exceed 30 percent when sodium sulfate is used or 38 percent when magnesium sulfate is used. For Rock Types 2 and 3, the average percent weight loss must not exceed 38 percent for sodium sulfate soundness or 45 percent for magnesium sulfate soundness.

Figure 523–1 Isoline Map of the Freeze-Thaw Severity Index for Contiguous 48 United States (map is from ASTM D5312)



D. Field Durability Inspection

- (1) Rock that fails to meet the material requirements stated above (if specified), may be accepted only if similar rock from the same source has been demonstrated to be sound after 5 years or more of service under conditions of weather, wetting and drying, and erosive forces similar to those anticipated for the rock to be installed under this specification.
- (2) A rock source may be rejected if the rock from that source deteriorates in less than 5 years under similar use and exposure conditions expected for the rock to be installed under this specification, even though it meets the testing requirements stated above.
- (3) Deterioration is defined as the loss of more than one-quarter of the original rock volume, or severe cracking that would cause a block to split. Measurements of deterioration are taken from linear or surface area particle counts to determine the percentage of deteriorated blocks. Deterioration of more than 25 percent of the pieces must be cause for rejection of rock from the source.

E. Grading

The rock must conform to the specified grading limits after it has been placed within the matrix of the rock riprap. Grading tests must be performed, as necessary, according to ASTM D5519, Method A, B, or C, as applicable.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 531—Portland Cement

A. Scope

This specification covers the quality of Portland cement.

B. Quality

- (1) Portland cement shall conform to the requirements of ASTM C150 for the specific types of cement such as Type I, IA, II, IIA, II(MH), II(MH)A, III, IIIA, IV, and V Portland cement.
- (2) Type IS Portland blast-furnace slag cement, Type IP Portland-pozzolan cement, or Type IL Portland-limestone cement shall conform to the requirements of ASTM C595 and may be used unless prohibited by the specifications.
- (3) When air-entraining cement is required, the contractor shall furnish the manufacturer's written statement providing the source, amount, and brand name of the air-entraining component.

C. Storage at the Construction Site

Cement shall be stored and always protected from weather, dampness, or other destructive elements. Cement that is partly hydrated or otherwise damaged will not be accepted.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 533—Chemical Admixtures for Concrete

A. Scope

This specification covers the quality of chemical admixtures for manufacturer of Portland cement concrete.

B. Quality

- (1) Air-entraining admixtures shall conform to the requirements of ASTM C260.
- (2) Water-reducing and/or retarding admixtures shall conform to the requirements of ASTM C494, Types A, B, D, F, or G.
- (3) Plasticizing or plasticizing and retarding admixtures shall conform to ASTM C494, Types F or G.
- (4) Accelerating or water-reducing and accelerating admixtures shall be noncorrosive and conform to the requirements of ASTM C494, Types C and E. The manufacturer shall provide long-term test data results from an independent laboratory verifying that the product is noncorrosive when used in concrete exposed to continuously moist conditions.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 534—Concrete Curing Compound

A. Scope

This specification covers the quality of liquid membrane-forming compounds suitable for spraying on concrete surfaces to retard the loss of water during the concrete curing process.

B. Quality

The curing compound must meet the requirements of either ASTM Specification C309 or C1315. If Type 1 is specified, a fugitive dye must be used.

C. Delivery and Storage

All curing compounds must be delivered to the site of the work in the original container bearing the name of the manufacturer and the brand name. The compound must be stored in a manner that prevents damage to the container and protects water-emulsion types from freezing.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 539—Steel Reinforcement (for concrete)

A. Scope

This specification covers the quality of steel reinforcement for reinforced concrete.

B. Quality

- (1) All reinforcement shall be free from loose or flaky rust, soil, oil, grease, paint, or other deleterious matter.
- (2) Steel bars for concrete reinforcement shall be grade 40, 50, or 60 deformed bars conforming to one of the following specifications:
 - (i) Deformed and plain billet-steel bars for concrete reinforcement—ASTM A615.
 - (ii) Rail-steel deformed bars for concrete reinforcement—ASTM A996.
 - (iii) Axle-steel deformed bars for concrete reinforcement—ASTM A996.
- (3) Dowels shall be plain round bars conforming to the same specifications listed above for steel bars.
- (4) Fabricated deformed steel bar mats for concrete reinforcement shall conform to the requirements of ASTM A184.
- (5) Deformed and plain steel welded wire reinforcement shall conform to the requirements of ASTM A1064.
- (6) Epoxy-coated steel bars for concrete reinforcement shall conform to the requirements of ASTM A775.

C. Dimensions of Welded Wire Reinforcement

Gauges, diameters, spacing, and arrangement of wires for welded steel wire fabric shall be as defined for the specified style designations.

D. Storage

Steel reinforcement inventories at the site of the work shall be stored above the ground surface on platforms, skids, or other supports and shall be kept clean and protected from mechanical injury and corrosion.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 547—Plastic Pipe

A. Scope

This specification covers the quality of Poly Vinyl Chloride (PVC), Polyethylene (PE), High Density Polyethylene (HDPE), and Acrylonitrile-Butadiene-Styrene (ABS) plastic pipe, fittings, and joint materials.

B. Material

- (1) Pipe—The pipe must be as uniform as commercially practicable in color, opaqueness, density, and other specified physical properties. It must be free from visible cracks, holes, foreign inclusions, or other defects. The dimensions of the pipe must be measured as prescribed in ASTM D2122. Unless otherwise specified, the pipe must conform to the requirements listed in this specification and the applicable reference specifications in Figure 547–2, the requirements specified in Construction Specification 45, Plastic Pipe, and the requirements shown on the drawings.
- (2) Fittings and joints—Fittings and joints must be of a schedule, SDR or DR, pressure class, external load carrying capacity, or pipe stiffness that equals or exceeds that of the plastic pipe. The dimensions of fittings and joints must be compatible with the pipe and measured in accordance with ASTM D2122. Joint and fitting material must be compatible with the pipe material. The joints and fittings must be as uniform as commercially practicable in color, opaqueness, density, and other specified physical properties. It must be free from visible cracks, holes, foreign inclusions, or other defects. Fittings and joints must conform to the requirements listed in this specification, the requirements of the applicable specification referenced in the ASTM or AWWA specification for the pipe, the requirements specified in Construction Specification 45, and the requirements shown on the drawings.
- (3) Solvents—Solvents for solvent-welded pipe joints must be compatible with the plastic pipe used and must conform to the requirements of the applicable specification referenced in the ASTM or AWWA specification for the pipe, fitting, or joint.
- (4) Gaskets—Rubber gaskets for pipe joints must conform to the requirements of ASTM F477, Elastomeric Seals (Gaskets) for Jointing Plastic Pipe.

C. Perforations

When perforated pipe is specified, perforations must conform to the following requirements unless otherwise specified in Construction Specification 45 or shown on the drawings:

- (1) Perforations must be either circular or slots.
- (2) Circular perforations must be $1/4 \pm 1/16$ -inch diameter holes arranged in rows parallel to the axis of the pipe.
- (3) Perforations must be evenly spaced along each row such that the center-to-center distance between perforations is not less than eight times the perforation diameter. Perforations may appear at the ends of short and random lengths. The minimum perforation opening per foot of pipe must be as shown in Figure 547–1.

Figure 547-1 Perforations

Nominal pipe size	Minimum number of rows		Minimum opening/foot (in ²)
	circular	slot	
4	2	2	0.22
6	4	2	0.44
8	4	2	0.44
10	4	2	0.44
12	6	2	0.66

- (4) Rows must be arranged in two equal groups at equal distance from the bottom on each side of the vertical centerline of the pipe. The lower-most rows of perforations must be separated by an arc of not less than 60 degrees or more than 125 degrees. The uppermost rows of perforations must be separated by an arc not to exceed 166 degrees. The spacing of rows between these limits must be uniform. The minimum number of rows must be as shown in Figure 547–1.
- (5) Slot perforations must be symmetrically located in two rows, one on each side of the pipe centerline. Slot perforations must be located within the lower quadrants of the pipe with slots no wider than 1/8 inch and spaced not to exceed 11 times the perforation width. Minimum perforation opening per lineal foot of pipe must be as shown in Figure 547–1.
- (6) On both the inside and outside of the pipe, perforations must be free of cuttings or frayed edges and of any material that would reduce the effective opening.

Figure 547-2 Pipe Specifications

Pipe	Specification
Poly vinyl chloride (PVC) pipe	
Plastic pipe - Schedules 40, 80, 120	ASTM D1785 ASTM D2466
Pressure rated pipe - SDR Series	AWWA C900 ASTM D2241
Plastic drain, waste, and vent pipe and fittings	ASTM D2665
Joints for IPS PVC pipe using solvent weld cement	ASTM D2672
Composite sewer pipe	ASTM D2680
Type PSM PVC sewer pipe and fittings	ASTM D3034
Large-diameter gravity sewer pipe and fittings	ASTM F679
Smooth-Wall Underdrain Systems for Highway, Airport, and Similar Drainage	ASTM F758
Profile gravity sewer pipe and fittings based on controlled inside diameter	ASTM F794
Corrugated sewer pipe with a smooth	ASTM F949

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Pipe	Specification
interior and fittings	
Pressure pipe, 4-inch through 60-inch for water distribution	AWWA C900
Polyethylene (PE) plastic pipe	
12- to 60-inch annular corrugated profile-wall polyethylene (PE) pipe and fittings	ASTM F2306
SIDR-PR based on controlled inside diameter	ASTM D2239
SDR-PR based on controlled outside diameter	ASTM D3035
High density polyethylene (HDPE) plastic pipe	
Plastic pipe and fittings	ASTM D3350
SDR-PR based on controlled outside diameter	ASTM F714
Heat joining polyolefin pipe and fittings	ASTM D2657
Acrylonitrile-butadiene-styrene (ABS) pipe	
Composite sewer pipe	ASTM D2680

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 592—Geotextile

A. Scope

This specification covers the quality of geotextile, including geotextile for temporary silt fence.

B. General Requirements

- (1) Fibers (threads and yarns) used in the manufacture of geotextile must consist of synthetic polymers composed of a minimum of 85 percent by weight polypropylene, polyester, polyamide, polyethylene, polyolefin, or polyvinylidene-chloride. The fiber must be formed into a stable network of filaments retaining dimensional stability relative to each other. The geotextile must be free of defects, such as holes, tears, and abrasions. The geotextile must be free of any chemical treatment or coating that significantly reduces its porosity. Fibers must contain stabilizers, inhibitors, or both to enhance resistance to ultraviolet light. Geotextile other than for temporary silt fence must conform to the requirements in Figure 592-1 or 592-2, as applicable. Geotextile for temporary silt fence must conform to ASTM D6461.
- (2) Thread used for factory or field sewing must be of a color contrasting to the color of the fabric and made of high-strength polypropylene, polyester, or polyamide material. It must be as resistant to ultraviolet light as the geotextile being sewn.

C. Classification

- (1) There are two geotextile classifications, woven and nonwoven. Geotextile for temporary silt fence may be either woven or nonwoven. Silt film woven geotextile may not be used except for temporary silt fence.
- (2) Woven geotextiles are made from fabric that is formed by the uniform and regular interweaving of the threads or yarns in two directions. Woven fabrics must be manufactured from monofilament yarn formed into a uniform pattern with distinct and measurable openings, retaining their position relative to each other. The fabric must have a selvage edge or otherwise be finished to prevent unraveling.
- (3) Nonwoven geotextiles are made from fabric that is formed by a random placement of threads in a mat and bonded by needle punching, heat bonding, or resin bonding. Nonwoven geotextiles must have distinct but variable small openings, retaining their position relative to each other when bonded. The use of heat- or resin-bonded nonwovens is restricted as specified in note 2 of Figure 592-2.

D. Sampling and Testing

The geotextile must conform to Figure 592-1 or 592-2 or ASTM D6461, as applicable, for the product type shown on the label. Documentation described in either (i) or (ii) below is required to verify the product meets the specified requirements:

- (i) Product properties as listed in the latest edition of the "Specifiers Guide," Geosynthetics (Industrial Fabrics Association International, 1801 County Road B, West Roseville, MN 55113-4061 or at <http://www.geosindex.com>), and that represent average roll values, are acceptable.

- (ii) Test data from the geotextile production run for each of the specified tests listed in Figure 592-1 or 592-2 or ASTM D6461, as applicable.

E. Shipping and Storage

Each roll of geotextile must be labeled or tagged to clearly identify the brand, class, and individual production run in accordance with ASTM D4873. The geotextile must be shipped and transported in rolls wrapped with a cover for protection from moisture, dust, dirt, debris, and ultraviolet light. The cover must be maintained undisturbed to the maximum extent possible before placement.

Figure 592-1 Requirements for Woven Geotextiles ^{1/}

Property	Test method	Class I	Class II	Class III	Class IV
Grab tensile strength (lb)	ASTM D4632	247 minimum	180 minimum	180 minimum	315
Elongation at failure (%)	ASTM D4632	<50	<50	<50	<50
Trapezoidal tear strength (lb)	ASTM D4533	90 minimum	67 minimum	67 minimum	112 minimum
Puncture strength (lb)	ASTM D6241	495 minimum	371 minimum	371 minimum	618 minimum
Ultraviolet stability (% retained strength)	ASTM D4355	50 minimum	50 minimum	50 minimum	70 minimum
Permittivity (sec ⁻¹)	ASTM D4491	as specified			
Apparent opening size (AOS) ^{2/}	ASTM D4751	as specified			
Percent open area (POA) (%)	USACE ^{3/} CWO-02215-86	as specified			

^{1/} All values are minimum average roll values (MARV) in the weakest principal direction, unless otherwise noted.

^{2/} Maximum average roll value.

^{3/} Note: CWO is a USACE reference.

Figure 592-2 Requirements for Non-Woven Geotextiles ^{1/}

Property	Test method	Class I ^{2/}	Class II ^{2/}	Class III ^{2/}	Class IV ^{2/}
Grab tensile strength (lb)	ASTM D4632 grab test	202 minimum	157 minimum	112 minimum	202 minimum
Elongation at failure (%)	ASTM D4632	50 minimum	50 minimum	50 minimum	50 minimum
Trapezoidal tear strength (lb)	ASTM D4533	79 minimum	56 minimum	40 minimum	79 minimum
Puncture strength (lb)	ASTM D6241	433 minimum	309 minimum	223 minimum	433 minimum
Ultraviolet light (% retained strength)	ASTM D4355	50 minimum	50 minimum	50 minimum	50 minimum
Permittivity sec ⁻¹	ASTM D4491	0.70 minimum or as specified			
Apparent opening size (AOS) (mm) ^{3/}	ASTM D4751	0.22 maximum or as specified			

^{1/} All values are minimum average roll values (MARV) in the weakest principal direction, unless otherwise noted.

^{2/} Needle punched geotextiles may be used for all classes. Heat-bonded or resin-bonded geotextiles may be used for class IV only.

^{3/} Maximum average roll value.